

Supplementary material for ‘Latent coverage from noisy calibration’

Kun Woo Park

Equation, theorem, table and section numbers without the prefix S refer to the main paper. The sections follow the order of the main paper, after a section of preliminaries. Each numerical statement says how it is supported: proved, certified in ball arithmetic, computed in double precision, or found by an exploratory search; § S5 keeps these levels apart, and § S11 records a conjecture that no result uses. Table S1 shows where each result of the main paper is supported.

Table S1: Where the results of the main paper are supported

main-paper result	supplement	support
Proposition 1 at both ends; one end is not enough	§ S2	proved
Theorem 1(i), symmetric unimodal noise	§ S2, Proposition S1	proved
Theorem 1(ii), mean-zero log-concave noise	§ S2	proved; constants certified in ball arithmetic
Theorem 1(iii), mean-zero unimodal noise	§ S2	proved
Proposition 2	§ S2	proved; examples in closed form
Theorem 2; ranks of Table 1	§ S3	proved; ranks computed exactly
Proposition 3; Table 2	main text; § S9	proved; exact binomial values and simulation
Theorem 3	§ S4; constants § S5	proved; c_q certified in ball arithmetic
feasibility bound and the 1.16% and 0.22% widenings	§ S4, feasibility; shape of the radius	proved; 0.22% in double precision
location of the mean (§ 4)	§ S4, Theorems S1 and S2	proved; κ_q^* in double precision
Corollary 1	§ S4, § S5	proved; slack intervals certified
rules with known variances (§ S6)	§§ S3, S4, S6	proved; computed radius in double precision
Table 3	§ S9	simulation, exact conditional coverage
shift of the new residual (§ 6)	§ S8	proved

S1 Preliminaries

Notation. The noise is e , and ϵ is a standardized noise variable, with law in $\mathcal{E}$; ε denotes a small positive number, and e is the base of the natural logarithm. Other symbols are defined where they are used.

Bi-log-concave laws. $\mathcal{B}$ is the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave (Dümbgen et al., 2017); $\mathcal{L} \subset \mathcal{B}$. The following facts are used throughout.

1. A law in $\mathcal{B}$ that is not a point mass has a positive density f on the interior of its support, the hazard $f/(1 - F)$ is nondecreasing, f/F is nonincreasing, and $f' \leq f^2/F$ where f is differentiable (Dümbgen et al., 2017).

2. $\mathcal{B}$ is closed under convolution with a log-concave law. If $\log F_X$ is concave and g is a log-concave density, $(t, y) \mapsto F_X(t - y)g(y)$ is log-concave on $\mathbb{R}^2$, so $F_{X+Y}(t) = \int F_X(t - y)g(y) dy$ is log-concave by the theorem of Prékopa (1973), and likewise $1 - F_{X+Y}$.
3. $\mathcal{B}$ is closed under weak convergence (Saumard, 2019): concavity of $\log F_n$ and $\log(1 - F_n)$ passes to the limit at continuity points, and a limit that is not a point mass has no atom, because a concave function is continuous on the interior of its domain.
4. If X has a continuous distribution function F with $\log F$ concave and a finite mean, then $F\{E(X)\} \geq e^{-1}$: $F(X)$ is uniform, so $E\{\log F(X)\} = -1$, and Jensen's inequality gives $\log F\{E(X)\} \geq E\{\log F(X)\}$. If also $\log(1 - F)$ is concave, $1 - F\{E(X)\} \geq e^{-1}$ in the same way.

S2 Coverage transfer for classes of noise laws

This section proves the results of § 2: the use of Proposition 1 at both ends of the interval, the three parts of Theorem 1, and Proposition 2.

Proposition 1 at both ends

Let $\lambda_{\mathcal{E}}(p') = \sup\{q' : \Psi_{\mathcal{E}}(q') < p'\}$, which is the inverse of $\Psi_{\mathcal{E}}$ when $\Psi_{\mathcal{E}}$ is continuous and increasing, and let $\phi_{\mathcal{E}}(a) = 1 - \lambda_{\mathcal{E}}(1 - a)$. Both are nondecreasing.

Let $W \in \mathcal{B}$, let the noise be $e = \sigma\epsilon$ with ϵ in $\mathcal{E}$ and $\sigma > 0$, and let $a_R = \text{pr}(W + e > 1)$ and $a_L = \text{pr}(W + e < -1)$. The law of $Y = W - 1$ has $\log F_Y$ concave and $\text{pr}(Y + e \leq 0) = 1 - a_R$, and e has a law in $\mathcal{E}$. By Proposition 1, $F_Y(0) \geq q'$ for every q' with $\Psi_{\mathcal{E}}(q') < 1 - a_R$, so $\text{pr}(W > 1) \leq \phi_{\mathcal{E}}(a_R)$. The law of $-W - 1$ has a concave log distribution function, because $\log(1 - F_W)$ is concave and F_W is continuous, and $-e$ has a law in $\mathcal{E}$; so $\text{pr}(W < -1) \leq \phi_{\mathcal{E}}(a_L)$. Hence, if $\phi_{\mathcal{E}}(a_L) + \phi_{\mathcal{E}}(a_R) \leq 1 - q$ whenever $a_L + a_R \leq 1 - p$, every $W \in \mathcal{B}$ with $\text{pr}(|W + e| \leq 1) \geq p$ has $\text{pr}(|W| \leq 1) \geq q$, for every noise law in $\mathcal{E}$ and every scale.

In the proof of Theorem 2 this bound is applied to one unit at a time, so the noise laws may differ between units. Since $\phi_{\mathcal{E}}$ is nondecreasing, it suffices to check finitely many splits (a_L, a_R) , as in § S5. A certificate replaces $\lambda_{\mathcal{E}}$ by a smaller certified level, which only enlarges $\phi_{\mathcal{E}}$ and so keeps the bound valid.

One end is not enough

No noisy level below $\Psi_{\mathcal{E}}(q)$ suffices. If $\Psi_{\mathcal{E}}(q) > p$, take ϵ in $\mathcal{E}$ with $E\{\min(1, qe^{-\epsilon})\} > p$, let $a = \log(1/q)$, and let $W = 1 + s(a + \delta - E_1)$ with noise $s\epsilon$. As $s \rightarrow 0$ the left end contributes nothing, so the noisy coverage of $[-1, 1]$ tends to $E\{\min(1, qe^{-\delta-\epsilon})\} > p$ for small $\delta > 0$, while the latent coverage tends to $qe^{-\delta} < q$.

Proof of Theorem 1(i): symmetric unimodal noise

By the theorem of Khintchine, a law symmetric and unimodal about 0 is that of RU , with U uniform on $[-1, 1]$ and $R \geq 0$ independent of U . Hence $E\{\min(1, qe^{-RU})\} = E\{g(R)\} \leq \sup_r g(r)$, with $g(r) = E\{\min(1, qe^{-rU})\}$, and uniform laws attain the supremum. With $\ell = \log q$,

$$g(r) = q \frac{\sinh r}{r} \quad (r \leq -\ell), \quad g(r) = \frac{\ell + r + 1 - qe^{-r}}{2r} \quad (r \geq -\ell).$$

The first branch increases. On the second, $g'(r)$ has the sign of $d(r) = qe^{-r}(1+r) - (1+\ell)$, which decreases, with $d(-\ell) = f(\ell) = e^{2\ell}(1-\ell) - 1 - \ell > 0$ because $f(0) = f'(0) = 0$ and $f''(\ell) = -4\ell e^{2\ell} > 0$ for $\ell < 0$, and $d(r) \rightarrow -(1+\ell) < 0$. So g is maximal at the root r of d , where $g(r) = \{1 + qe^{-r}\}/2$.

Heavy tails are allowed: the t_3 and Cauchy laws are symmetric and unimodal, and at $q = 0.9$ the supremum over the scale is 0.90013 for t_3 , and for the Cauchy law it is 0.9, approached as the scale tends to 0 but not attained (numerical).

Symmetric unimodal noise: the exact two-sided level

Proposition S1. *For symmetric unimodal noise, $\Psi = \Psi_{\mathcal{E}}$ is increasing and strictly convex on $(e^{-1}, 1)$, and for $1/2 < p < 1$ and $e^{-1} < q < 1$ the two-sided level is $p_{\mathcal{E}}(q) = \Psi(q)$: noisy coverage p guarantees latent coverage $\lambda(p) = \Psi^{-1}(p)$ for every $W \in \mathcal{B}$ and every symmetric unimodal noise, possibly different between units, and no larger latent level is guaranteed, even for log-concave W .*

Proof. Write $s = qe^{-r}$ with r the root of d , so that $\Psi(q) = (1 + s)/2$ and $s(1 + r) = 1 + \ell$. Since $\partial d / \partial r = -qre^{-r} < 0$, r is differentiable in q ; differentiating $s(1 + r) = 1 + \ell$ with $ds = s(dq/q - dr)$ gives $rs dr/dq = \ell/q$ and, using $\ell = s(1 + r) - 1$, $ds/dq = (rs - \ell)/(qr) = (1 - s)/(qr)$. Hence $\Psi'(q) = (1 - s)/(2qr) > 0$ and, after the same substitution,

$$\Psi''(q) = \frac{(1 - s)\{1 - s(1 + r)^2\}}{2q^2r^3s}, \quad 1 - s(1 + r)^2 = 1 - (1 + \ell)(1 + r).$$

Let $x = 1 + \ell \in (0, 1)$ and $R = 1/x - 1$. As d decreases on $(0, \infty)$, $r < R$ if and only if $d(R) < 0$, that is $\ell - R < 2 \log x$, that is $k(x) = 2 \log x - x + 1/x > 0$; and $k(1) = 0$, $k'(x) = -(1 - x)^2/x^2 < 0$. So $(1 + \ell)(1 + r) < 1$ and $\Psi'' > 0$. As $q \rightarrow 1$, $0 < r < R \rightarrow 0$ and $s \rightarrow 1$, so Ψ extends continuously with $\Psi(1) = 1$, and $\Psi(q) \rightarrow 1/2$ as $q \downarrow e^{-1}$.

The inverse λ is therefore increasing and concave on $[1/2, 1]$, and $\phi(a) = 1 - \lambda(1 - a)$ is nondecreasing and convex on $[0, 1/2]$ with $\phi(0) = 0$, hence superadditive: $\phi(a_L) + \phi(a_R) \leq \phi(a_L + a_R)$. By Proposition 1, letting the latent level increase to $\lambda(1 - a)$, an end with noisy failure probability a has latent failure probability at most $\phi(a)$; the left end uses the negated noise, again symmetric unimodal. With $a_L + a_R \leq 1 - p$ the latent failure probability is at most $\phi(1 - p) = 1 - \lambda(p)$, which is at most $1 - q$ when $p \geq \Psi(q)$.

Conversely, let $p < \Psi(q)$ and let ϵ be uniform on $[-r, r]$, with r the maximizer. By the converse in Proposition 1 there is an exponential tail $Y = a + \eta - E_1$, $\eta > 0$ small, with $F_Y(0) < q$ and $\text{pr}(Y + \epsilon \leq 0) > p$. For $\sigma > 0$ put $W = t + \sigma Y$, which is log-concave, and noise $\sigma \epsilon$. Then $\text{pr}(|W| \leq t) \leq F_Y(0) < q$, while $\text{pr}(|W + \sigma \epsilon| \leq t) = \text{pr}(Y + \epsilon \leq 0) - \text{pr}(Y + \epsilon < -2t/\sigma)$, which exceeds p for small σ , since ϵ is bounded. The same construction with $q' > \lambda(p)$, for which $\Psi(q') > p$, shows that $\lambda(p)$ cannot be raised. $\square$

At $q = 0.8, 0.9$ and 0.95 the two-sided levels are therefore $\Psi(q) = 0.8077245, 0.9017873$ and 0.9504311 (ball arithmetic, enclosures of width below 10^{-15}), and noisy coverage 0.9 guarantees exactly latent coverage $\lambda(0.9) = 0.8981432$, with $\Psi(0.8981) < 0.89996$ and $\Psi(0.8982) > 0.90005$. The same argument applies to any class $\mathcal{E}$ closed under scaling and negation whose one-sided function $\Psi_{\mathcal{E}}$ is convex; for Gaussian and mean-zero log-concave noise convexity is not proved, and the two-sided levels rest on the split certificates.

The split certificate of § S5, run for this class as an independent check (`split_certificate_su`), gives $p_{\mathcal{E}} \leq 0.8078, 0.9018$ and 0.9505 at $q = 0.8, 0.9$ and 0.95 , in agreement with Proposition S1.

Proof of Theorem 1(ii): mean-zero log-concave noise

Reduction and parametrization. Let $h(y) = \min(1, qe^{-y})$, which is continuous, nonincreasing and 1-Lipschitz. If ϵ is log-concave with mean zero, conditioning on $|\epsilon| \leq M$ and recentring changes $E\{h(\epsilon)\}$ by at most $2\text{pr}(|\epsilon| > M) + |E(\epsilon \mid |\epsilon| \leq M)|$, which tends to zero; so it suffices to bound $E\{h(\epsilon)\}$ over log-concave laws on compact intervals with $E(\epsilon) \geq 0$, a larger class. By the localization theorem (Fradelizi & Guédon, 2004) the supremum of this linear functional is attained at a point mass or at a law with density proportional to $e^{\beta y}$ on a segment. A law with $E(\epsilon) > 0$ can be translated to mean zero without decreasing

$E\{h(\epsilon)\}$, because h is nonincreasing. So the supremum is over the point mass at 0, where $h(0) = q$, and the mean-zero log-affine laws on segments. Write the latter as $\epsilon = L(V - \mu_k)$, V on $[0, 1]$ with density proportional to e^{kv} , $k = \beta L$, $\mu_k = E(V)$; then

$$E\{h(\epsilon)\} = \frac{1}{Z_k} \left\{ \int_0^{v_c} e^{kv} dv + qe^{L\mu_k} \int_{v_c}^1 e^{(k-L)v} dv \right\}, \quad v_c = \min\{1, \max(0, \mu_k + \ell/L)\},$$

with $Z_k = \int_0^1 e^{kv} dv$. For $|k| \geq 2$ we use the scale $\sigma = L/|k|$ and the truncated exponential form $\epsilon = \pm\sigma(\tau - m)$, τ standard exponential truncated to $[0, |k|]$ and $m = E(\tau)$, which includes $|k| = \infty$, the centred exponential laws $\pm\sigma(E - 1)$. As $|k| \geq 2$, m lies between $m_2 = 1 - 2/(e^2 - 1)$ and 1.

Upper bound on a box. A branch and bound in Arb ball arithmetic, `uai.noise_classes.lc_certificate_parallel`, bounds $E\{h(\epsilon)\}$ from above on boxes of (k, L) for $|k| \leq 2$ and of $(1/|k|, \sigma)$ for $|k| \geq 2$, splitting until each box is below the target. On a box the bound is

$$E\{h(\epsilon)\} \leq J(u) = \text{pr}(\epsilon \leq u) + qE\{e^{-\epsilon}; \epsilon > u\},$$

which holds for every cut u because $h(y) \leq 1(y \leq u) + qe^{-y}1(y > u)$. The cut is fixed at the kink of h for the box centre, so that it does not depend on the parameters within the box; since J' vanishes at the kink, the bound is tight to second order. Near $k = 0$, Z_k and $\int_0^1 ve^{kv} dv$ are evaluated by their series with a remainder bound. Box ends enter as exact doubles, and derived ends such as $1/|k|$ are rounded outward before they enter the ball computations.

Ends of the scale parameter. The ends of the scale parameter are closed analytically.

1. For small L , $E\{h(\epsilon)\} \leq qE(e^{-\epsilon}) \leq qe^{L^2/24}$. With $K(s) = \log \int_0^1 e^{sv} dv$ and $\epsilon = L(V - \mu_k)$, $\log E(e^{-\epsilon}) = K(k - L) - K(k) + LK'(k) = \int_{k-L}^k (s - k + L)K''(s) ds \leq L^2/24$, because $K''(s) = s^{-2} - \{2 \sinh(s/2)\}^{-2}$, the variance of the law on $[0, 1]$ with density proportional to e^{sv} , is at most its value $1/12$ at $s = 0$, as $x^{-2} - \sinh^{-2} x$ decreases in $|x|$. This improves on the bound $qe^{L^2/8}$ of Hoeffding's lemma.
2. For large L , $h(y) \leq 1$ for $y \leq 4$ and $h(y) \leq qe^{-4}$ for $y > 4$, so $E\{h(\epsilon)\} \leq \text{pr}(\epsilon \leq 4) + qe^{-4}$. Here $\text{pr}(\epsilon \leq 4) = \text{pr}(V \leq \mu_k + 4/L) \leq 1 - e^{-1} + 4d/L$, by fact 4 of § S1 applied to V and because the density of V is at most $d = 2/(1 - e^{-2})$ for $|k| \leq 2$. The resulting bound $1 - e^{-1} + 4d/L + qe^{-4}$ is below the target once $L \geq L_{\text{big}}$, which is found by enlarging L until the inequality holds in ball arithmetic.
3. For the scale σ of the truncated exponential form, with $\text{pr}(\tau \leq y) \leq (1 - e^{-y})/(1 - e^{-2})$ and $\text{pr}(\tau \geq y) \leq e^{-y}/(1 - e^{-2})$ as $|k| \geq 2$:
 - $\epsilon = \sigma(\tau - m)$, small σ : $E\{h(\epsilon)\} \leq qe^{\sigma m}E(e^{-\sigma\tau}) \leq qe^{\sigma}$;
 - $\epsilon = \sigma(\tau - m)$, large σ : $E\{h(\epsilon)\} \leq \text{pr}(\tau \leq 1 + 4/\sigma) + qe^{-4}$;
 - $\epsilon = \sigma(m - \tau)$, $\sigma < 1$: $E\{h(\epsilon)\} \leq qe^{-\sigma m}E(e^{\sigma\tau} | \tau \leq |k|) \leq qe^{-\sigma m_2}/(1 - \sigma)$;
 - $\epsilon = \sigma(m - \tau)$, large σ : $E\{h(\epsilon)\} \leq \text{pr}(\tau \geq m_2 - 4/\sigma) + qe^{-4}$.

Each bound is below the target for σ beyond a threshold at its end, found in ball arithmetic as for L .

Results. At $q = 0.9$ the certificate gives $\Psi_{\mathcal{E}}(0.9) < 0.9052$; boxes are split until their bounds fall below this target, so the largest bound, just below 0.9052, reflects the stopping rule; the target itself lies 2.4×10^{-5} above the value at the centred exponential law. The centred exponential with $\sigma = 0.11777$ gives $\Psi_{\mathcal{E}}(0.9) > 0.90517$; it gives $\Psi_{\mathcal{E}}(0.8942) > 0.9$ at $\sigma = 0.1259$.

For the two-sided levels, a partition of the noisy failure probability $a_L \in [0, (1 - p)/2]$ at the left end is proposed from the centred exponential values: on each interval $[a_0, a_1]$ the bound $\phi_{\mathcal{E}}(a_1) + \phi_{\mathcal{E}}(1 - p - a_0)$,

evaluated at those values, uses at most half of the margin below $1 - q$, and the other half is left for the certificates. Every node a is then certified by the branch and bound at a level q_a below $\lambda_{\mathcal{E}}(1 - a)$; this certifies the noisy level 0.906 for latent level 0.9, with 41 intervals, and noisy level 0.9 for latent level 0.8935, with 45 intervals (`split_certificate_lc`).

Numerically the worst split puts the whole failure probability at one end, as for Gaussian and symmetric unimodal noise, so the two-sided level would be $\Psi_{\mathcal{E}}(0.9) = 0.905176$ (numerical). Numerically, along truncated exponential laws $\sigma(\tau - m)$ with a long right tail the supremum over the scale increases with the truncation length, to the centred exponential; for the reflected laws it decreases. These two statements are numerical and are not used.

Proof of Theorem 1(iii): mean-zero unimodal noise

Let ϵ have density $(1 - \rho)/a$ on $[-a, 0]$ and ρ/b on $(0, b]$ with $b = (1 - \rho)a/\rho$ and $\rho \leq 1/2$; it is unimodal with mean zero. Then $E\{h(\epsilon)\} \geq (1 - \rho)(1 + \ell/a)$, which tends to one as $\rho \rightarrow 0$ and $a \rightarrow \infty$, so $\Psi_{\mathcal{E}}(q) = 1$ and no noisy level below one transfers to any latent level.

Gaussian noise as a class

For Gaussian noise $\Psi_{\mathcal{E}}(q) = \sup_{\sigma} E\{\min(1, qe^{-\sigma Z})\}$, and Proposition 1 recovers the sign of the constants of Lemma S1: $c_{p,q} \leq 0$ if $\Psi_{\mathcal{E}}(q) < p$ and $c_{p,q} > 0$ if $\Psi_{\mathcal{E}}(q) > p$. At $q = 0.9$ the supremum is 0.900803, at $\sigma = 0.0503$, in the slack interval $(0.90079, 0.901]$ of Corollary 1.

Proof of Proposition 2: unimodal latent laws

The bound needs no shape. Given $W > 1$, the event $\epsilon \geq 0$ gives $W + \epsilon > 1$, and given $W < -1$ the event $\epsilon \leq 0$ gives $W + \epsilon < -1$; both have probability at least $\beta_{\mathcal{E}}$, since $\mathcal{E}$ is closed under negation, so $1 - p \geq \text{pr}(|W + \epsilon| > 1) \geq \beta_{\mathcal{E}}\text{pr}(|W| > 1)$. For median-zero noise this is the bound of Guille-Escuret & Ndiaye (2024); for mean-zero log-concave noise $\text{pr}(\epsilon \geq 0) \geq e^{-1}$ and $\text{pr}(\epsilon \leq 0) \geq e^{-1}$ by fact 4 of § S1, with equality for the centred exponential law $E_1 - 1$; for mean-zero unimodal noise $\beta_{\mathcal{E}} = 0$.

For the construction let ϵ have a law in $\mathcal{E}$, continuous at 0, with $\text{pr}(\epsilon \geq 0) = \beta < \beta_{\mathcal{E}} + \gamma'$; as $p > 1 - \beta_{\mathcal{E}}$, $(1 - p)/\beta < 1$ for small γ' . Let $\tau, \gamma'' > 0$ be small and $0 < \delta < 1$; δ will tend to 0 with σ , and $s/\delta \geq m/2$ once δ is small. Let

$$s = (1 - p - \gamma'' - \tau)/\beta, \quad m = 1 - s - \tau, \quad A = 2s(1 + \delta/2)/\tau - 1,$$

so that $s, m > 0$, and let W have a density made of three pieces:

1. $m/2$ on $[-1, 1]$, the mass inside the interval;
2. s/δ on $(1, 1 + \delta]$, a thin layer just outside the right end;
3. $\tau/(A - 1)$ on $[-A, -1)$, a long light left tail that makes $E(W) = s(1 + \delta/2) - \tau(A + 1)/2 = 0$.

For small τ the density is nondecreasing up to $1 + \delta$ and zero beyond it, so W is unimodal. With noise $\sigma\epsilon$, the failure probability of $|W + \sigma\epsilon| \leq 1$ splits over the three pieces:

1. inside, $(m/2) \int_{-1}^1 \text{pr}(|w + \sigma\epsilon| > 1) dw$, which tends to 0 as $\sigma \rightarrow 0$ by dominated convergence;
2. the layer, at most $s\{\text{pr}(\epsilon > -\delta/\sigma) + \text{pr}(\epsilon < -2/\sigma)\}$; the second term, a large negative noise carrying the point past -1 , tends to 0, and the first tends to β as $\sigma \rightarrow 0$ with $\delta = \sigma^2$, so that $\delta/\sigma \rightarrow 0$, by continuity at 0;

3. the left tail, at most τ .

Limits. The noisy coverage therefore tends to at least $1 - \tau - s\beta = p + \gamma'' > p$, while the latent coverage is $m = 1 - (1 - p - \gamma'' - \tau)/\beta - \tau$, within any γ of $1 - (1 - p)/\beta_{\mathcal{E}}$ for small γ' , γ'' and τ .

If $\beta_{\mathcal{E}} = 0$, take $\beta < \gamma'$ and $s = 1 - \tau - \gamma$, so that the latent coverage is $m = \gamma$ while the noisy coverage tends to at least $1 - \tau - s\beta \geq 1 - \tau - \gamma' > p$ for small τ and γ' .

The noisy coverage of this law has a closed form: for a piece with uniform density on $[a, b]$, $\int_a^b F(c - w) dw = G(c - a) - G(c - b)$, where F is the distribution function of the noise and G an antiderivative, $G(y) = y\Phi(y/\sigma) + \sigma\phi(y/\sigma)$ for Gaussian noise. With $\tau = 10^{-4}$, $\delta = 10^{-3}\sigma$ and the noisy coverage set to 0.9, the latent coverage is 0.870, 0.820, 0.807 and 0.801 at $\sigma = 0.1, 0.03, 0.01$ and 0.001 for Gaussian noise, 0.876, 0.822, 0.807 and 0.801 for uniform noise of standard deviation σ , and 0.809, 0.751, 0.736 and 0.729 for the centred exponential noise $\sigma(E_1 - 1)$, against the limits 0.8, 0.8 and $1 - e/10 = 0.72817 \dots$ (experiments/e42_unimodal_boundary.py).

S3 Finite-sample ranks

This section gives the computations behind Theorem 2 and Table 1, and the order-statistic level used by the rules with known variances.

Smallest valid ranks

By Theorem 2 the smallest valid rank is at most the smallest k with $\text{pr}\{\text{Bin}(K, p) \leq k - 1\} \geq 1 - \delta$ at $p = p_{\mathcal{E}}(q)$, and at least the smallest such k at $p = \Psi_{\mathcal{E}}(q)$. At $q = 0.9$ and $\delta = 0.05$ a rank exists only if $1 - p^K \geq 0.95$, that is for $K \geq 29$ under Gaussian and symmetric unimodal noise, $K \geq 31$ under mean-zero log-concave noise and $K \geq 59$ at $p = 0.95$. Against the usual rank, with $p = 0.9$, the raise for Gaussian noise is 0 at $K = 110$, 1 at $K = 300$ and 10^3 , 3 at $K = 3000$ and 8 to 10 at $K = 10^4$. The function `certified_rank` returns this rank.

Unimodal latent laws. The proof of Theorem 2 uses only the transfer for one unit. With the transfer of Proposition 2, which needs noisy level $1 - \beta_{\mathcal{E}}(1 - q)$, that is $(1 + q)/2$ for median-zero noise and $1 - (1 - q)/e$ for mean-zero log-concave noise, it gives the ranks in the last two rows of Table 1: if $\text{pr}(|W| \leq s) < q$ then $\text{pr}(|V_i| \leq s) < 1 - \beta_{\mathcal{E}}(1 - q)$, for every latent law. These ranks are the smallest valid ones for unimodal latent laws with mean zero. Let $p < 1 - \beta_{\mathcal{E}}(1 - q)$ and take the construction in the proof of Proposition 2 with the layer mass s chosen in $1 - q - \tau < s < (1 - \tau - p)/\beta$, which is possible for small τ and γ' . Then the latent coverage of $[-1, 1]$ is $m = 1 - s - \tau < q$, so $r = Q_q(|W|) > 1$, and the noisy coverage of $[-1, 1]$ tends to at least $1 - \tau - s\beta > p$, so $\text{pr}(|V_i| < r) \geq \text{pr}(|V_i| \leq 1) > p$ for small σ . As in the proof of Theorem 2, the reliability of rank k is then at most $\text{pr}\{\text{Bin}(K, p) \leq k - 1\}$.

Proposition 3 is illustrated at an extremal law in § S9, with exact reliabilities up to $K = 10^6$.

An order-statistic level for known variances

The computed radius and the shrunk threshold of § S6 use a level for the average noisy distribution function rather than a bound for each unit. Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ -quantile of the Beta($k, K + 1 - k$) distribution.

Proposition S2. *If $k \geq Kp_k + 1$ then $\text{pr}\{\bar{F}(T) \geq p_k\} \geq 1 - \delta$.*

Proof of Proposition S2. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \geq k$. N is a sum of independent indicators whose success probabilities average p_k , and $k - 1 \geq Kp_k$. By Hoeffding (1956), $\text{pr}(N \leq k - 1)$ is smallest when all success probabilities are equal. Hence $\text{pr}(N \geq k) \leq \text{pr}\{\text{Bin}(K, p_k) \geq k\} = \text{pr}\{\text{Beta}(k, K + 1 - k) \leq p_k\} = \delta$. $\square$

S4 Gaussian noise: the sharp radius and the location of the mean

This section proves the results of § 4 and states and proves Theorems S1 and S2 on the location of the mean, for Gaussian noise $e \sim N(0, D)$, after scaling the threshold t to 1, with $x = D/t^2$ and $R_{p,q}$ defined by (2).

Reduction to log-affine laws

Write $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1)$, so that the constraint in (2) reads $E\{g_x(W)\} \geq p$, and let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $e^{\beta w}$ on a segment.

Proposition S3. $R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F}, E\{g_x(W)\} \geq p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § S6.

The proof uses the localization theorem; over $\mathcal{F}$ the problem is a two-dimensional maximization of closed forms.

Proof of Proposition S3. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E\{g_x(W)\} \geq p$ and let $s < Q_q(|W|)$, so that $\text{pr}(|W| < s) \leq q - \gamma$ for some $\gamma > 0$. Choose $M \geq s$ with $g_x(w) \leq p$ for $|w| \geq M$, and let W_M be W conditioned on $|W| \leq M$, which is log-concave. With $\varepsilon_M = \text{pr}(|W| > M)$, the conditioning removes only mass on which the kernel is at most p , so

$$E\{g_x(W_M)\} \geq \frac{p - p\varepsilon_M}{1 - \varepsilon_M} = p, \quad \text{pr}(|W_M| < s) \leq \frac{q - \gamma}{1 - \varepsilon_M} < q$$

for M large. On $[-M, M]$ the constraint function $g_x - p$ is continuous, and $\mu \mapsto -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence because $(-s, s)$ is open. By the localization theorem (Kannan et al., 1995; Fradelizi & Guédon, 2004), the supremum of this functional over the log-concave probability measures on $[-M, M]$ that satisfy $\int (g_x - p) d\mu \geq 0$ is attained at a point mass or at a law with log-affine density on a segment. Hence some member of $\mathcal{F}$ is feasible and has $\text{pr}(|W| < s) < q$, that is, $Q_q(|W|) \geq s$. Since $s < Q_q(|W|)$ was arbitrary the claim follows. Only continuity of the kernel and its decay at infinity were used, so the argument applies to the average kernel $\bar{g}$ of § S6. $\square$

For $\beta > 0$ and $\sigma = x^{1/2}$, integration by parts gives

$$\int_a^b e^{\beta w} \Phi\left(\frac{c-w}{\sigma}\right) dw = \frac{1}{\beta} \left[e^{\beta w} \Phi\left(\frac{c-w}{\sigma}\right) \right]_a^b + \frac{e^{\beta c + \beta^2 \sigma^2 / 2}}{\beta} \left\{ \Phi\left(\frac{b-c-\beta\sigma^2}{\sigma}\right) - \Phi\left(\frac{a-c-\beta\sigma^2}{\sigma}\right) \right\}, \quad (\text{S1})$$

so the constraint is a combination of normal distribution functions. For fixed slope and length, the translations c of a member of $\mathcal{F}$ that satisfy the constraint form an interval, because $c \mapsto \text{pr}(|W + c + e| \leq 1)$ is log-concave (Prékopa, 1973), and $c \mapsto \text{pr}(|W + c| \leq s)$ is log-concave, so $c \mapsto Q_q(|W + c|)$ is quasi-convex and is maximized at an end of that interval. Each end is found by a root-finding in c , and the quantile by a second root-finding, using (S1).

The one-sided problem

Near each end of the interval the problem is one-sided. For $u > 0$ let E_u be exponential with rate u , independent of Z , and for the exponential tail $b - E_u$ let

$$m_u(b) = \text{pr}(b - E_u + Z \leq 0) = \Phi(-b) + e^{-ub + u^2/2} \Phi(b - u),$$

which is continuous and strictly decreasing in b . Let $b_u(p)$ solve $m_u(b) = p$, and write $v_q(u) = b_u(q) - u^{-1} \log(1/q)$ and $k_q(u) = u^{-1} - b_u(q)$; the law $b_u(q) - E_u$ has q -quantile $v_q(u)$ and mean $-k_q(u)$.

Lemma S1. Let $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L}, \text{pr}(Y + Z \leq 0) \geq p\}$ and $c_q = c_{q,q}$. Then $c_{p,q} = \sup_{u>0}\{b_u(p) - u^{-1} \log(1/q)\}$, $c_q > 0$, and $q \mapsto c_q$ is nonincreasing. The same holds with $\mathcal{B}$ in place of $\mathcal{L}$.

Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$ (§ S5). The proofs of Lemma S1 and of Lemma S2 below replace Y by an exponential tail through a supergradient of $\log F_Y$ at its q -quantile, which keeps the quantile, the constraint and the bound on the mean.

Proof of Lemma S1. Exponential tails are log-concave, which gives one inequality. Conversely, let $Y \in \mathcal{L}$ with $\text{pr}(Y + Z \leq 0) \geq p$. If $Y = y_0$ is a point mass then $y_0 \leq -\Phi^{-1}(p) = \lim_{u \rightarrow \infty} \{b_u(p) - u^{-1} \log(1/q)\}$. Otherwise $F = F_Y$ is continuous and $\log F$ is concave (Prékopa, 1973). Let $v = F^{-1}(q)$ and $G(y) = F(v+y)$. The point 0 is interior to $\{G > 0\}$, so the right derivative $u \geq 0$ of $\log G$ at 0 is a supergradient, and $G(y) \leq qe^{uy}$ for all y . If $u = 0$ then $G \leq q < 1$, which is impossible, so $u > 0$ and $G(y) \leq \min(qe^{uy}, 1)$, which is the distribution function of $b_0 - E_u$ with $b_0 = u^{-1} \log(1/q)$. Hence $Y' = v + b_0 - E_u$ lies stochastically below Y and has q -quantile v , and

$$p \leq \text{pr}(Y + Z \leq 0) \leq \text{pr}(Y' + Z \leq 0) = m_u(v + b_0).$$

Since m_u is decreasing, $v \leq b_u(p) - b_0$. For monotonicity, let $q' > q$, let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$, and put $\rho = (q' - q)/(1 - q)$. Then Y conditioned on $Y \geq F_Y^{-1}(\rho)$ is log-concave; over $\mathcal{B}$ it stays in $\mathcal{B}$, because $\log(F - \rho) = h(\log F)$ with $h(z) = \log(e^z - \rho)$ nondecreasing and concave, and the log survival function of the conditioned law is $\min\{0, \log(1 - F) - \log(1 - \rho)\}$, a minimum of concave functions. It satisfies the constraint at level $(q' - \rho)/(1 - \rho) = q$, because the removed mass has noisy constraint value at most one, and has q -quantile v . Point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \leq c_q$. Finally, $c_q > 0$ because $v_q(u) = u^{-1} \log\{e^{u^2/2} \Phi(b_u - u) + e^{ub_u} \Phi(-b_u)\} = u/2 + o(u)$ as $u \rightarrow 0$, where $b_u = b_u(q)$ and $ub_u \rightarrow \log(1/q)$. $\square$

Theorem 3: the lower bound

Proof of Theorem 3, lower bound. Fix $\varepsilon > 0$ and choose u with $v_q(u) > c_q - \varepsilon$. For $p' \in [q, (1 + q)/2]$, $b_u(p') \geq b_{\min} > -\infty$. Let $Y = b_u(p') - E_u$ and $W = 1 + x^{1/2}Y$. With $\tau = 2x^{-1/2}$,

$$\text{pr}(|W + x^{1/2}Z| \leq 1) = p' - \text{pr}(Y + Z < -\tau) \geq p' - \eta_x, \quad \eta_x = e^{-u(\tau + b_{\min})/2} + \Phi\{-(\tau + b_{\min})/2\},$$

by splitting the event $Y + Z < -\tau$ according to whether $Z < -(\tau + b_{\min})/2$. Take $p' = q + \eta_x$, which lies in $[q, (1 + q)/2]$ for small x . Since $\text{pr}(|W| \leq s) \leq F_W(s)$, $Q_q(|W|) \geq 1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1} \log(1/q)\}$. As $\eta_x \rightarrow 0$ and b_u is continuous, $\liminf_{x \rightarrow 0} \{R_{q,q}(x) - 1\}/x^{1/2} \geq c_q - \varepsilon$. $\square$

Feasibility and the size of the widening

The kernel $g_x(w) = \text{pr}(|w + x^{1/2}Z| \leq 1)$ is symmetric in w and maximal at $w = 0$, so $E\{g_x(W)\} \leq g_x(0) = 2\Phi(x^{-1/2}) - 1$ for every W . A latent law is therefore feasible at level q only if $2\Phi(x^{-1/2}) - 1 \geq q$, that is $x^{1/2} \leq 1/\Phi^{-1}\{(1+q)/2\}$. With Theorem 3, the widening $Q_q(|W|) - 1 \leq c_q x^{1/2}$ is at most $c_q/\Phi^{-1}\{(1+q)/2\}$, which is $0.01906/1.6449 < 1.16\%$ at $q = 0.9$ for every $W \in \mathcal{B}$. The smaller bound 0.22% for log-concave laws rests on the double-precision bounds of § S5 (§ S4).

The location of the mean

Let $R_{p,q}^{(b)}$ be defined as $R_{p,q}$ in (2) with the additional constraint $|E(W)| \leq b$, so that the mean is at least $1 - b$ from ± 1 . A mean away from the boundary reduces the order of the widening from $x^{1/2}$ to x .

Theorem S1. Let $W \in \mathcal{B}$ with $|E(W)| \leq M$, let $\sigma > 0$ and $t > M$, and suppose that $\text{pr}(|W + \sigma Z| \leq t) \geq q$. Then

$$Q_q(|W|) \leq M + \{(t - M)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - M)}.$$

In particular $R_{p,q}^{(b)}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$.

Here $\sigma^2 = D$ in the original units. A mean far from an end of the interval caps how steeply the density can rise towards it; the proof makes this precise with the heat equation for quantiles (Steinbrecher & Shaw, 2008). The order x cannot be improved, even for mean-zero laws: centred log-affine laws attain it (Proposition S4), so centring does not replace symmetry.

The order $x^{1/2}$ returns only when the mean is within a bounded number of noise standard deviations of the boundary. With $b_x = 1 - \kappa x^{1/2}$, the constraint $|E(W)| \leq b_x$ places the mean at least κ noise standard deviations, $\kappa D^{1/2}$, inside the nearer end of the interval. Let

$$L_q(\kappa) = \sup\{F_Y^{-1}(q) : Y \in \mathcal{B}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\}.$$

Theorem S2. Let $1 - e^{-1} < q < 1$, $x > 0$ and $W \in \mathcal{B}$ with $\text{pr}(|W + x^{1/2}Z| \leq 1) \geq q$. Then $|E(W)| < 1$, and if $\Delta = Q_q(|W|) - 1 > 0$,

$$2\{1 - |E(W)|\}\Delta + \Delta^2 \leq x. \quad (\text{S2})$$

Moreover, for every $\kappa \geq 0$,

$$\lim_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa),$$

and the same limit holds with $\mathcal{L}$ replaced by $\mathcal{B}$ in the definition of $R^{(b)}$.

Inequality (S2) is Theorem S1 with $t = 1$; at $1 - |E(W)| = \kappa x^{1/2}$ it gives $\Delta \leq \{(\kappa^2 + 1)^{1/2} - \kappa\}x^{1/2}$, and the limit gives the exact constant. The condition $q > 1 - e^{-1}$ gives $|E(W)| < 1$, since a law in $\mathcal{B}$ has at least mass e^{-1} on each side of its mean. The function L_q equals c_q up to $\kappa \approx 20$ at $q = 0.9$ and then decays like $1/\kappa$ (Lemma S2).

Lemma S2. Let $e^{-1} < q < 1$ and $\kappa \geq 0$. Then

$$\sup\{F_Y^{-1}(q) : Y \in \mathcal{B}, E(Y) \leq -\kappa, \text{pr}(Y + Z \leq 0) \geq q\} = L_q(\kappa) = \sup\{v_q(u) : u > 0, k_q(u) \geq \kappa\}.$$

The function L_q is positive and nonincreasing, and, for $q > 1/2$, $L_q(\kappa) = c_q$ for $\kappa \leq \kappa_q^*$, where κ_q^* is the largest value of k_q on the compact set of maximizers of v_q , and $\kappa L_q(\kappa) \rightarrow \{1 - \log(1/q)\}/2$ as $\kappa \rightarrow \infty$. In double precision $\kappa_q^* = 6.81, 20.18$ and 50.07 at $q = 0.8, 0.9$ and 0.95 .

Proof of Lemma S2. For $q > 1/2$, v_q is continuous on $(0, \infty)$, tends to 0 as $u \rightarrow 0$ and to $-\Phi^{-1}(q) < 0$ as $u \rightarrow \infty$, and is positive somewhere, so its maximizers form a nonempty compact subset of $(0, \infty)$ and κ_q^* is well defined. Let $a = \log(1/q)$ and $A = 1 - a > 0$.

Replacement by an exponential tail. Let Y be nondegenerate with q -quantile v . Since $\log F_Y$ is concave, a supergradient $u > 0$ at v gives $F_Y(y) \leq \min\{qe^{u(y-v)}, 1\}$, the distribution function of $Y' = v + a/u - E_u$; so Y' lies stochastically below Y and has the same q -quantile. Then $E(Y') \leq E(Y) \leq -\kappa$ and $\text{pr}(Y' + Z \leq 0) \geq q$, so the replacement keeps both constraints.

Which rates are feasible. For the tail $b - E_u$ the constraints read $b \leq b_u(q)$ and $b \leq u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A/u - \kappa\}$. If $k_q(u) < \kappa$ this minimum is $A/u - \kappa$; since k_q is continuous and $k_q(s) \rightarrow \infty$ as $s \rightarrow 0$, some $s < u$ has $k_q(s) = \kappa$, where the value is $A/s - \kappa > A/u - \kappa$. So only u with $k_q(u) \geq \kappa$ matter, and there the value is $v_q(u)$. A point mass y_0 satisfying the constraints has $y_0 = E(Y) \leq -\kappa \leq 0 < L_q(\kappa)$, where positivity holds because $v_q(u) \sim u/2$ and $k_q(u) \sim A/u$ as $u \rightarrow 0$.

Large κ . Since $m_u(b) \geq \Phi(-b)$, $b_u(q) \geq -\Phi^{-1}(q)$, so k_q is bounded on $[u_0, \infty)$ for every $u_0 > 0$ and $k_q(u) \geq \kappa$ forces $u \rightarrow 0$ as $\kappa \rightarrow \infty$; moreover $ub_u(q) \rightarrow \log(1/q)$, so $k_q(u) = \{A + o(1)\}/u$, and $v_q(u) = u/2 + o(u)$ as $u \rightarrow 0$ (proof of Lemma S1). These expansions give $\kappa L_q(\kappa) \rightarrow A/2$: for large κ only small u are feasible, $u \leq \{A + o(1)\}/\kappa$, and $v_q(u) \leq \{1 + o(1)\}u/2$ with equality at the boundary. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. $\square$

Proof of Theorem S1. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq M$. Otherwise, for $s > 0$ the law of $W + s^{1/2}Z$ is in $\mathcal{B}$ (Saumard, 2019), with mean μ , distribution function F_s and a positive smooth density f_s . For $r > \mu$ let $h = f_s(r)/F_s(r)$. The tangent of $\log F_s$ at r gives $F_s(y) \leq \min\{F_s(r)e^{h(y-r)}, 1\}$, the distribution function of an exponential tail with mean $r - \{1 + \log F_s(r)\}/h$ that lies stochastically below $W + s^{1/2}Z$, so $\mu \geq r - \{1 + \log F_s(r)\}/h$ and $h(r - \mu) \leq 1$. Laws in $\mathcal{B}$ satisfy $f'_s \leq f_s^2/F_s$ (Dümbgen et al., 2017), so $(\log f_s)'(r) \leq h \leq 1/(r - \mu)$; applied to $-W$ this gives $(\log f_s)'(-r) \geq -1/(r + \mu)$. Hence, for $r > M$,

$$f'_s(r) \leq \frac{f_s(r)}{r - M}, \quad -f'_s(-r) \leq \frac{f_s(-r)}{r - M}.$$

Let $H_s(r) = \text{pr}(|W + s^{1/2}Z| \leq r)$. By the heat equation $\partial_s f_s = f''_s/2$,

$$\partial_s H_s(r) = \frac{1}{2}\{f'_s(r) - f'_s(-r)\} \leq \frac{f_s(r) + f_s(-r)}{2(r - M)} = \frac{\partial_r H_s(r)}{2(r - M)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \geq -1/[2\{r(s) - M\}]$ where $r(s) > M$. Hence $\{r(s) - M\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\varepsilon) - M\}_+^2 \leq \{r(\sigma^2) - M\}_+^2 + \sigma^2 - \varepsilon$. As $\varepsilon \rightarrow 0$, $r(\varepsilon) \rightarrow Q_q(|W|)$, because a nondegenerate law in $\mathcal{B}$ has a positive density on the interior of its support (Dümbgen et al., 2017), so the distribution function of $|W|$ is continuous and strictly increasing where it lies strictly between 0 and 1. Since $r(\sigma^2) \leq t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \leq d + \sigma^2/(2d)$. $\square$

Theorem S2: the transition

Proof of Theorem S2, first part. Let $V = W + x^{1/2}Z$ with distribution function F . Since $\log F$ and $\log(1 - F)$ are concave and $F(V)$ is uniform, $E\{\log F(V)\} = -1$ and Jensen's inequality gives $F\{E(V)\} \geq e^{-1}$ and likewise $1 - F\{E(V)\} \geq e^{-1}$. If $E(W) \geq 1$ then $\text{pr}(|V| \leq 1) \leq F\{E(W)\} \leq 1 - e^{-1} < q$, and similarly if $E(W) \leq -1$; so $|E(W)| < 1$. Theorem S1 with $M = |E(W)|$ and $t = 1$ gives $\Delta \leq (d^2 + x)^{1/2} - d$ with $d = 1 - |E(W)|$, which is (S2). $\square$

Proof of Theorem S2, the limit. Lower bound. For the lower bound in the limit, take u with $k_q(u) \geq \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - E_u\}$ of the lower bound in Theorem 3. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \leq 1 - \kappa x^{1/2}$, because b_u decreases in its argument, and it is positive for small x ; hence $\liminf_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \geq v_q(u)$ for every such u .

The upper bound is proved by contradiction in four steps.

Step 1: rescaling near the boundary. For the upper bound, suppose that $x_n \rightarrow 0$, $h_n = x_n^{1/2}$, and that W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$ with $Q_q(|W_n|) - 1 \geq a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (S2), $\kappa h_n \leq 1 - |E(W_n)| \leq \bar{\kappa} h_n$ with $\bar{\kappa} = (1 - a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \geq 0$; let $Y_n = (W_n - 1)/h_n$, so that $-\bar{\kappa} \leq E(Y_n) \leq -\kappa$.

Step 2: a variance bound. The law of $T_n = Y_n + Z$ is in $\mathcal{B}$, with a smooth positive density of maximum M_n and distribution function F_n , and $F_n(0) \geq \text{pr}(|W_n + h_n Z| \leq 1) \geq q$ while $F_n\{E(T_n)\} \leq 1 - e^{-1}$ by the first part. So $q - 1 + e^{-1} \leq F_n(0) - F_n\{E(T_n)\} \leq M_n \bar{\kappa}$. For a law in $\mathcal{B}$ with smooth density g , maximum M and median m , the hazard $g/(1 - F)$ is nondecreasing, so $g(y) \leq 2g(m)\{1 - F(y)\} \leq 2g(m)$ for $y \leq m$, and g/F is nonincreasing, which gives the same for $y \geq m$; so $g(m) \geq M/2$; the tangent bounds

for $\log F$ and $\log(1 - F)$ at m give $\text{pr}(|T - m| > y) \leq e^{-2g(m)y}$, hence $\text{var}(T) \leq 1/\{2g(m)^2\} \leq 2/M^2$. Thus $\text{var}(Y_n) \leq 2\bar{k}^2/(q - 1 + e^{-1})^2$, the Y_n are tight and uniformly integrable.

Step 3: the limit. Along a subsequence $Y_n \rightarrow Y$ in distribution with $Y \in \mathcal{B}$, possibly a point mass, since $\mathcal{B}$ is closed under weak convergence (Saumard, 2019), $E(Y) \leq -\kappa$ and $\text{pr}(Y + Z \leq 0) = E\{\Phi(-Y)\} \geq q$.

Step 4: contradiction with the one-sided optimum. On $\{W_n \geq 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so $a_0 \leq \{Q_q(|W_n|) - 1\}/h_n \rightarrow Q_q(Y) \leq L_q(\kappa)$ by Lemma S2, where the quantile converges because a law in $\mathcal{B}$ has a unique q -quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$, so $\limsup_{x \rightarrow 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} \leq a_0$ for every $a_0 \in (L_q(\kappa), 1)$; if $L_q(\kappa) \geq 1$ there is nothing to prove, because (S2) gives $\Delta < x^{1/2}$. $\square$

Centred laws

Proposition S4. *Let X have density proportional to $e^{\beta x}$ on $[-\ell, 0]$ with $\beta > 0$ and $0 < \ell \leq \infty$, let $W = X - E(X)$ and let $r_0 = Q_q(|W|)$. If $\pm r_0$ lie in the interior of the support of W , then, after scaling W so that the noisy q -quantile of $|W + x^{1/2}Z|$ equals 1,*

$$Q_q(|W|) = 1 + \frac{1}{2}\beta r_0 \tanh(\beta r_0)x + o(x) \quad (x \rightarrow 0).$$

For every $q \in (0, 1)$ the condition holds for small $\beta\ell$, so $R_{q,q}^{(0)}(x) - 1$ is of exact order x .

Numerically the best coefficient in this family is 0.3466 at $q = 0.8$, for $W = 1 - E_1$, and 0.0747 at $q = 0.9$.

Proof of Proposition S4. Near $\pm r_0$ the density f of W is smooth with $f' = \beta f$, and the ends of the support are at positive distance. The jumps of the density at the ends of the support contribute $O\{v^{1/2}\phi(d_0/v^{1/2})\}$ near $\pm r_0$, where $d_0 > 0$ is the distance from $\pm r_0$ to the ends, which is $o(v)$. Hence $H_v(r) = \text{pr}(|W + v^{1/2}Z| \leq r) = H_0(r) + v\{f'(r) - f'(-r)\}/2 + o(v)$ uniformly near r_0 , and the noisy quantile is $t(v) = r_0 - v\beta\{f(r_0) - f(-r_0)\}/[2\{f(r_0) + f(-r_0)\}] + o(v) = r_0 - (\beta v/2)\tanh(\beta r_0) + o(v)$. Scaling by $t(v)$ and writing $x = v/t(v)^2$, a continuous and increasing function of v near 0, gives the expansion. As $\beta\ell \rightarrow 0$, W/ℓ tends to the uniform law on $[-1/2, 1/2]$, whose q -quantile of $|W|/\ell$ is $q/2 < 1/2$. $\square$

The case $p > q$

The constant of § 4.3 is

$$C_{p,q} = \sup_{\alpha_L + \alpha_R = 1-p} \inf_{\beta_L + \beta_R = 1-q} \max(c_{1-\alpha_L, 1-\beta_L}, c_{1-\alpha_R, 1-\beta_R}) \geq c_{p,q}, \quad (\text{S3})$$

where α_L, α_R split the noisy failure probability and β_L, β_R the latent one. If $C_{p,q} \leq 0$ then $R_{p,q}^{\mathcal{B}}(x) \leq 1$ for every x , and if $c_{p,q} > 0$ then $R_{p,q}(x) > 1$ for all small x ; Corollary 1 follows from the certified values $C_{0.901, 0.9} \leq 0$ and $c_{0.90079, 0.9} > 0$ of § S5. In $C_{p,q}$, $c_{1,q'} = -\infty$, since no law has noisy coverage one at a single end. Numerically the worst split puts the whole budget on one end, so that $C_{p,q} = c_{p,q}$; this observation is not used. Ball arithmetic gives $-0.0572 \leq C_{0.9036, 0.9} \leq -0.057$, where 0.9036 is the order-statistic level for $K = 110, k = 105$ at $\delta = 0.04$, used in § S6.

In the proof of the upper bound of Theorem 3, replace $1 - q$ by $1 - p$ for the noisy budget, choose any split $\beta_L + \beta_R = 1 - q$ of the latent budget, and use $F_Y^{-1}(1 - \beta_R) \leq c_{1-\alpha_R, 1-\beta_R}$ at each end. Since $c_{p', q'}$ decreases in p' , the worst case has $\alpha_L + \alpha_R = 1 - p$, which gives $R_{p,q}(x) \leq 1 + C_{p,q}x^{1/2}$ for every x , with $C_{p,q}$ nonincreasing in p . The proof of the lower bound applies with $b_u(q)$ replaced by $b_u(p)$ and $p' = p + \eta_x$, giving $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} - o(x^{1/2})$ for every $p \geq q$. This proves Corollary 1, whose numerical intervals are certified in § S5.

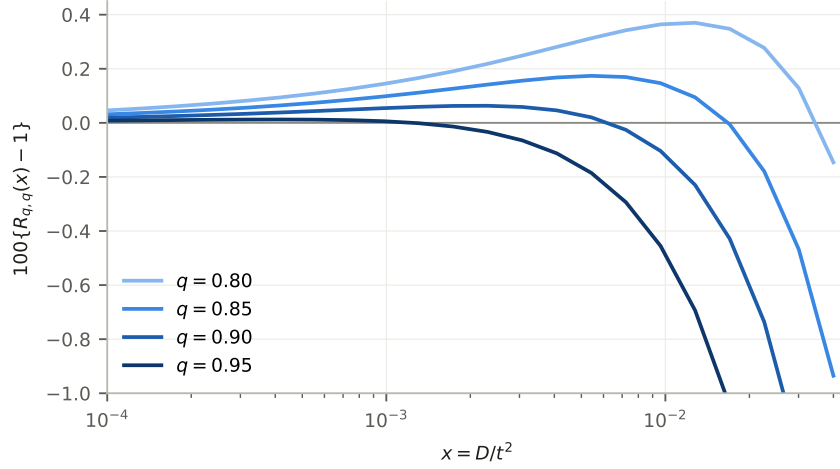


Figure S1: Sharp latent radius relative to the noisy threshold, $100\{R_{q,q}(x) - 1\}$, for four coverage levels, computed by grid maximization over the extremal family, which bounds $R_{q,q}$ from below.

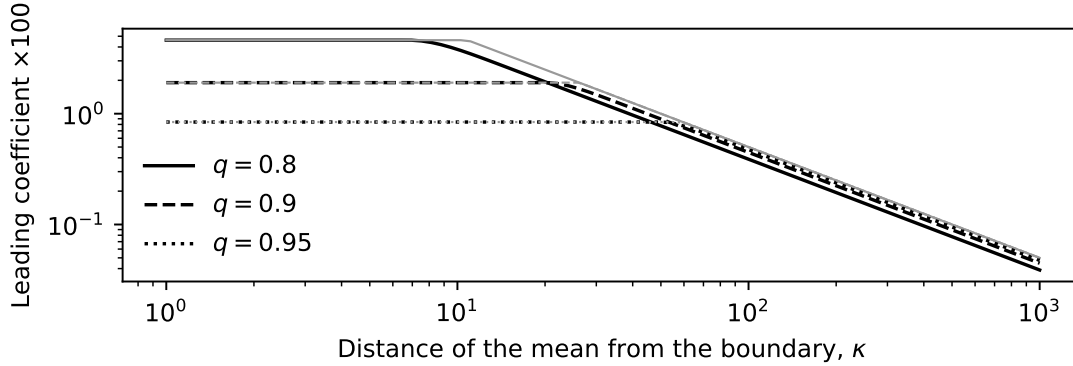


Figure S2: Leading coefficient $L_q(\kappa)$ of the worst-case radius correction, times 100, when the mean lies at distance at least $\kappa x^{1/2}$ from the boundary (Theorem S2), for $q = 0.8$ (solid), 0.9 (dashes) and 0.95 (dots), with the upper bound $\min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ from Theorems 3 and S1 (grey). Both axes are logarithmic.

The shape of the sharp radius

Figure S1 shows $R_{q,q}(x) - 1$ computed by grid maximization over the extremal family; grid values bound $R_{q,q}$ from below. The levels at which the grid value crosses 1 are $x^*(q) \approx 0.034, 0.017, 0.0062$ and 0.0012 for $q = 0.8, 0.85, 0.9$ and 0.95 , and the largest widening found is 0.37% , 0.17% , 0.063% and 0.012% . At $q = 0.9$ the numerical upper bounds of § S5 give $R_{q,q}(x) \leq 1$ for $x \geq 0.013$, so Theorem 3 bounds the widening by $c_{0.9} 0.013^{1/2} < 0.22\%$ at every x ; this rests on the double-precision bounds, whereas the bound 1.16% of § 4 rests only on Theorem 3 and the certified $c_{0.9}$. Below $x^*(q)$ the noisy threshold must be widened; above it, grid values below 1 do not by themselves show that shrinking is safe, which the numerical upper bounds of § S5 do. The exact grid values lie below $1 + c_{0.9}x^{1/2}$ at $x = 10^{-6}, \dots, 10^{-2}$, as Theorem 3 requires.

The transition: values

The following values are computed in double precision by `experiments/e32_centering.py` and are not certified. Table S2 lists, for $q = 0.8, 0.9$ and 0.95 , the end κ_q^* of the flat part of L_q and the maximizer u^* of v_q at which it is attained, the value of κ at which the two upper bounds in Fig. S2 cross, where L_q/U_q is smallest, with $U_q(\kappa) = \min\{c_q, (\kappa^2 + 1)^{1/2} - \kappa\}$ the upper bound from Theorems 3 and S1, that smallest ratio, and the limit $1 - \log(1/q)$ of the ratio as $\kappa \rightarrow \infty$.

Table S2: Transition values, double precision (`experiments/e32_centering.py`)

	$q = 0.8$	$q = 0.9$	$q = 0.95$
κ_q^*	6.8118	20.1823	50.0741
u^*	0.11328	0.044286	0.018943
crossing of the upper bounds, κ	10.829	26.221	59.466
smallest L_q/U_q	0.770	0.879	0.927
limit $1 - \log(1/q)$	0.777	0.895	0.949

At $q = 0.9$, $L_q(25) = 0.017393$ and $L_q(50) = 0.0089448$. For the mean-zero laws of Proposition S4, the coefficient with $\beta = 0.2$ and $\ell = 2$ at $q = 0.9$ is 0.0156540 ; the expansion is accurate to 6×10^{-5} at $x = 1.3 \times 10^{-4}$ but fails at $x \approx 1.1 \times 10^{-3}$, where the noise reaches the end of the support and the threshold may already be shrunk. As a numerical check of the proved bound of Theorem S1, including its form with $|E(W)| \leq M$, it was evaluated on 642 laws from the family of Proposition S4, with no violation.

The location of the mean in examples

Figure S3 shows the radius correction for laws scaled so that the noisy coverage of $[-1, 1]$ is exactly 0.9, computed from closed forms by `experiments/e35_mean_location.py`. For the boundary-layer law $W = 1 + x^{1/2}\{b - E_u\}$ at the maximizer u^* of v_q , whose mean lies $20.18 x^{1/2}$ from the boundary, the correction divided by $x^{1/2}$ is 0.019062 for $x \leq 3 \times 10^{-5}$, equal to $c_{0.9}$. For the same family at the rate with $k_q(u) = 50$ it is $0.008945 = L_{0.9}(50)$. For the fixed mean-zero law of Proposition S4 with $\beta\ell = 0.5$ on a unit segment, the correction divided by x is 0.024005 up to $x = 5 \times 10^{-5}$, and it turns negative once the noise reaches the end of the support. Larger centred coefficients sit closer to that end: at $\beta\ell = 0.9$ the coefficient is 0.070 but the quantile lies 0.0011 from the end, and the correction is positive only for x below about 10^{-7} . A Gaussian law with mean $1.29 x^{1/2}$ from the boundary after scaling, $W \sim N(1 - 2x^{1/2}, 0.01x)$ before scaling, needs no correction at all: its radius is $1.16 x^{1/2}$ below the noisy threshold. A mean near the boundary is necessary for a correction of order $x^{1/2}$, not sufficient.

The limit with κ fixed

Theorem S2 does not extend to κ growing like $x^{-1/2}$, that is, to centred laws; this subsection shows why. Applying the small- u construction to a centred law, that is, $\kappa = x^{-1/2}$, gives in the limit a left exponential tail of rate $1 - \log(1/q)$ whose noise-free central coverage is $q\{1 - e^{-2+2\log(1/q)}\}$, 0.750 at $q = 0.9$, far below the constraint. So $\{1 - \log(1/q)\}/2$ is not a lower bound for the centred coefficient.

Bi-log-concave laws: scope and examples

Lemma S1 holds over $\mathcal{B}$ with the same $c_{p,q}$, and so do the upper bounds of Theorem 3 and Corollary 1. The proof of Lemma S1 uses only the concavity of $\log F_Y$ at the q -quantile, so its upper bound holds for every Y with $\log F_Y$ concave, and exponential tails are in $\mathcal{L}$, which gives equality. In the proof of Theorem 3,

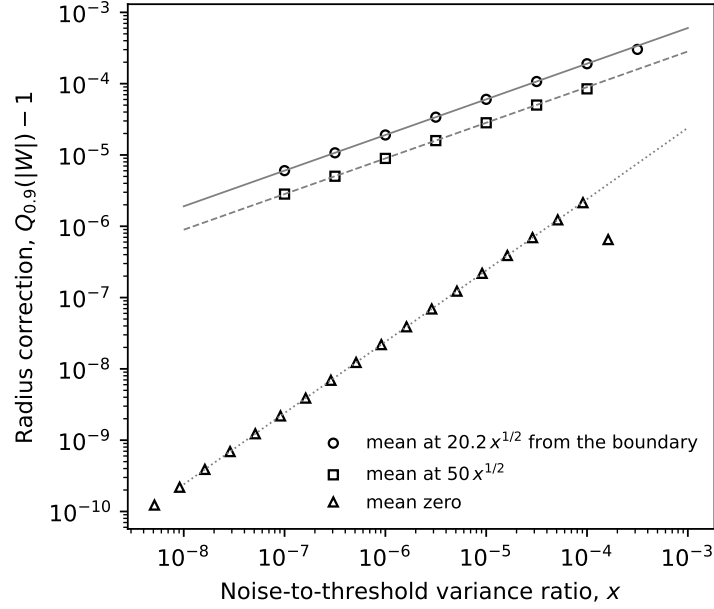


Figure S3: Radius correction $Q_{0.9}(|W|) - 1$ against x on logarithmic axes, for laws with noisy coverage exactly 0.9: mean at $20.2x^{1/2}$ from the boundary (circles), at $50x^{1/2}$ (squares) and mean zero (triangles), with $c_{0.9}x^{1/2}$ (solid), $L_{0.9}(50)x^{1/2}$ (dashes) and $0.024x$ (dots). Only positive corrections are shown.

$Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave, and $(-W - 1)/x^{1/2}$ has a concave log distribution function because $\log(1 - F_W)$ is concave and F_W is continuous. The lower bound uses laws in $\mathcal{L} \subset \mathcal{B}$.

Proposition S3 and the double-precision upper bounds of § S5 use log-concavity of the density; the ball-arithmetic constants do not. Theorems S1 and S2, Lemma S2, Theorem 2 and Corollary S1 hold over $\mathcal{B}$, by the facts of § S1: monotone hazards, the bound $f' \leq f^2/F$, and closure under convolution with a log-concave law and under weak limits.

Symmetry alone does not make noisy calibration conservative in $\mathcal{B}$. The density $f(w) = \{1 + 0.05 \cos(2\pi w)\}/2$ on $(-1, 1)$ is in $\mathcal{B}$, because $\sup |f'| = 0.05\pi < \inf f^2 = 0.2256$ (Dümbgen et al., 2017, Theorem 1), but it is not unimodal. At $q = 0.7$, $r_0 = Q_q(|W|) \approx 0.7077$ and $f'(r_0) > 0$; since f' is odd, $\partial_s \text{pr}(|W + s^{1/2}Z| \leq r_0) = f'(r_0) > 0$ at $s = 0$, so noise raises the coverage and the noisy quantile falls below r_0 , to 0.707676 at $s = 10^{-5}$, against $r_0 = 0.707678$. Anderson's comparison needs unimodality. The bound of Theorem S1 with $E(W) = 0$ holds for this cosine density, recentred to mean zero, and for its version with the cosine phase-shifted by $\pi/2$, at $q = 0.3, 0.7, 0.9$ and $s = 10^{-3}, 0.1, 1$ (test suite).

The bimodal law of Table S5, the mixture $\pi N(m_1, s^2) + (1 - \pi)N(m_2, s^2)$ with mean zero, variance one and $(m_2 - m_1)/s = 2.5$, is in $\mathcal{B}$ by the following lemma. With $\pi = 0.4$ the mixture stays in $\mathcal{B}$ up to a separation of about 2.59, while bimodality starts at about 2.443 (numerical). The parameters are $\pi = 0.4$, $m_1 = -0.94868$, $m_2 = 0.63246$ and $s = 0.63246$.

Lemma S3. *Let f be the density of $\pi N(m_1, s^2) + (1 - \pi)N(m_2, s^2)$ with $m_1 < m_2$ and $r = (m_2 - m_1)/s > 2$, let $P(w)$ be the posterior probability of the first component, and let $a < b$ solve $P(a) = \frac{1}{2} + h$ and $P(b) = \frac{1}{2} - h$, $h = (1/4 - 1/r^2)^{1/2}$. The mixture is in $\mathcal{B}$ if $f'F < f^2$ on $[a, m_2]$ and $-f'(1 - F) < f^2$ on $[m_1, b]$.*

Proof. Since $(\log f)'' = -s^{-2} + P(1 - P)r^2s^{-2}$ and P decreases, f is log-concave on $(-\infty, a]$ and on $[b, \infty)$. On $(-\infty, a]$, $F(t) = \int 1(y \leq t)f(y)1(y \leq a) dy$ is the marginal of a log-concave function of (t, y) , so $\log F$ is concave there (Prékopa, 1973); likewise $\log(1 - F)$ on $[b, \infty)$. Where $f' \leq 0$, in particular on $[m_2, \infty)$,

$f'F \leq f^2$ is trivial, and where $f' \geq 0$, in particular on $(-\infty, m_1]$, so is $-f'(1 - F) \leq f^2$. With the two hypotheses, $(\log F)'' \leq 0$ and $(\log(1 - F))'' \leq 0$ on the whole line. $\square$

For the law of Table S5 the two hypotheses are proved in Arb ball arithmetic on 128 boxes, with the ends a and b rounded outward (`uai.interval.bimodal.blc.certificate`, test suite); the largest values of $(f'F - f^2)/f^2$ on $[a, m_2]$ and of $\{-f'(1 - F) - f^2\}/f^2$ on $[m_1, b]$ are -0.141 and -0.864 .

S5 Numerical support

Three levels of numerical support are used and kept apart. Certificates in Arb ball arithmetic are proofs. Upper bounds evaluated in double precision with a margin are not interval arithmetic and are labelled numerical. Exploratory searches only propose values and are never used as bounds.

Certificates in ball arithmetic

Every deciding inequality is evaluated in Arb ball arithmetic; floating point only proposes points, balls are converted to doubles with outward rounding, and sums of bounds are formed as balls.

1. $m_u(b)$ decreases in b , and in u because E_u is stochastically decreasing in its rate. Hence $b_u(p)$ is nonincreasing in u , and on $[u_0, u_1]$, $b_u(p) - u^{-1} \log(1/q) \leq b_{u_0}(p) - u_1^{-1} \log(1/q)$. A ball with $m_u(b) < p$ proves $b_u(p) < b$.
2. For $u \geq u_{\max}$ the value is at most $b_{u_{\max}}(p)$. For small u , let $b(u) = \{\log(1/p) + u^2/2 - \log(1 - u^3)\}/u$. Then $e^{-ub+u^2/2} = p(1 - u^3)$, and the Mills ratio gives $\Phi(-b) \leq pu^3$ whenever $\phi\{\log(1/p)/u\} \leq p u^2 \log(1/p)$. This holds on $(0, u_{\min}]$ once it holds at $u_{\min} \leq \log(1/p)/2^{1/2}$, because the left side divided by u^2 increases there. So $b_u(p) \leq b(u)$, and for $p \geq q$ the resulting bound increases in u .
3. By Lemma S1, $c_{p',q'} \leq \gamma$ if and only if $1 - q' \geq \varphi_\gamma(1 - p')$, where $\varphi_\gamma(a) = \max[0, \sup_u 1 - \exp\{-u(b_u(1 - a) - \gamma)\}]$. Hence $C_{p,q} \leq \gamma$ if $\varphi_\gamma(a_L) + \varphi_\gamma(a_R) \leq 1 - q$ whenever $a_L + a_R = 1 - p$. Since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of a_L , with φ_γ bounded above by the same monotone branch and bound in u .

The certificates give $c_{0.8} \in [0.0460730, 0.0460731]$, $c_{0.9} \in [0.0190618, 0.0190619]$, $c_{0.95} \in [0.0084075, 0.0084077]$ and $c_{0.99} \in [0.0013950, 0.0013958]$, $-0.0571845 \leq c_{0.9036,0.9} \leq C_{0.9036,0.9} \leq -0.057$, $-0.1149813 \leq c_{0.9068,0.9} \leq C_{0.9068,0.9} \leq -0.114$, and $C_{p,q} \leq 0$ for $(q, p - q) = (0.8, 0.005)$, $(0.9, 0.001)$ and $(0.95, 0.0002)$. Lower bounds $c_{p,q} > 0$ at $p - q = 0.0043$, 0.00079 and 0.00015 come from single points u , evaluated in ball arithmetic, and $C_{p,q} \geq c_{p,q}$. Hence the smallest slack in Corollary 1 lies in $(4.3 \times 10^{-3}, 5.0 \times 10^{-3}]$, $(7.9 \times 10^{-4}, 1.0 \times 10^{-3}]$ and $(1.5 \times 10^{-4}, 2.0 \times 10^{-4}]$ at $q = 0.8, 0.9$ and 0.95 .

For the coverage form of Corollary 1, the same certificates give $C_{0.8,0.795} \leq 0$, $C_{0.9,0.8991} \leq 0$ and $C_{0.95,0.9498} \leq 0$, and enclosures give $c_{0.8,0.7954} \geq 7.0 \times 10^{-4}$, $c_{0.9,0.8992} \geq 4.2 \times 10^{-4}$ and $c_{0.95,0.9499} \geq 3.0 \times 10^{-3}$, so by the lower bound of the case $p \geq q$, $R_{p,q'}(x) > 1$ for small x at these pairs (`experiments/e37-coverage.transfer.py`).

Double-precision root finding puts the smallest slack at 4.4×10^{-3} , 8.0×10^{-4} and 1.6×10^{-4} . In coverage terms: as $x \rightarrow 0$ the boundary-layer laws $1 + x^{1/2}\{b_u(q) - E_u\}$, whose noisy coverage tends to exactly q , have latent coverage tending to $e^{-ub_u(q)}$, whose smallest values are 0.79532 , 0.89918 and 0.94984 at $q = 0.8, 0.9$ and 0.95 , attained at $u = 0.140$, 0.0508 and 0.0210 (double precision). Calibrating at the noisy level q thus costs at most a few thousandths of latent coverage in these examples, and the certified slack removes the loss. These last values are numerical.

The certificates for the noise classes in § S2 and for Lemma S3 follow the same conventions.

Numerical upper bounds of the radius in double precision

Code and file names that contain ‘certified’ for these quantities, such as `certified.R.csv`, `hetldc_certified` and `CertifiedShrinkTable`, are historical: the values are double-precision upper bounds, not ball-arithmetic certificates.

By the symmetry $W \mapsto -W$ of the kernel and of $|W|$ it suffices to treat $\beta \geq 0$. Write a member of $\mathcal{F}$ with $\beta \geq 0$ as $W = b - Y$, where Y has density proportional to $e^{-\beta y}$ on $[0, \ell]$. The ends of the family are included: $\ell = 0$ or $\beta = \infty$ gives a point mass and $\ell = \infty$ an exponential tail. For slopes $\beta' > \beta$ the likelihood ratio $e^{-(\beta' - \beta)y}$ is decreasing, so $Y_{\beta'}$ lies below Y_{β} in likelihood-ratio order, and for lengths $\ell < \ell'$ truncation gives $F_{\ell} \geq F_{\ell'}$. Thus W is stochastically increasing in β and b and decreasing in ℓ . On a box of parameters let W_{lo} and W_{hi} be the stochastically smallest and largest corners. Since $\text{pr}(W + e \leq \pm 1)$ and $\text{pr}(W \leq y)$ are expectations of decreasing functions of W , every law in the box satisfies

$$\text{pr}(|W + e| \leq 1) \leq \text{pr}(W_{\text{lo}} + e \leq 1) - \text{pr}(W_{\text{hi}} + e \leq -1), \quad \text{pr}(|W| \leq s) \geq \text{pr}(W_{\text{hi}} \leq s) - \text{pr}(W_{\text{lo}} \leq -s).$$

The density bound $\text{pr}(|W + e| \leq 1) \leq 2 \sup f_W = 2\beta/(1 - e^{-\beta\ell})$ is also available. A box whose first bound is below p , or whose second bound is at least q , contains no law that violates $R_{p,q}(x) \leq s$, and a branch and bound over boxes certifies $R_{p,q}(x) \leq s$. The location is confined to $[-1 + x^{1/2}\Phi^{-1}(p), b_{\max}]$. The lower end holds because $W \leq b$. For the upper end let $k_0 = 7x^{1/2}$. The density of W increases towards b , so $\text{pr}(|W| \leq 1 + k_0) \leq \rho_0/(1 + \rho_0)$ with $\rho_0 = (2 + 2k_0)/(b - 1 - k_0)$, and $\text{pr}(|e| > k_0) = 2\Phi(-7)$; $b_{\max}$ is the smallest b with $\rho_0/(1 + \rho_0) + 2\Phi(-7) < p$, beyond which no law is feasible. The parameters $\beta, \ell \in [0, \infty]$ are compactified.

Proposition S5. *For all $x, h \geq 0$, $R_{p,q}(x + h) \leq R_{p,q}(x) + c_q h^{1/2}$.*

Proof. If W is feasible at noise level $x + h$, then $W + h^{1/2}Z'$, with $Z' \sim N(0, 1)$ independent of (W, Z) , is log-concave and feasible at level x . So $s = Q_q(|W + h^{1/2}Z'|) \leq R_{p,q}(x)$, and Theorem 3 at scale s gives $Q_q(|W|) \leq s + c_q h^{1/2}$. $\square$

We computed numerical upper bounds of $R_{p,0.9}$ for $p = 0.9, 0.9036$ and 0.9068 on $0.002 \leq x \leq 0.366$ with step 5×10^{-4} , using Proposition S5 between grid points. Here 0.9068 and 0.9036 are the order-statistic levels p_k of $K = 110$, $k = 105$ at $\delta = 0.05$ and $\delta = 0.04$; the latter is used in § S6.

These bounds rest on double-precision evaluation of the closed forms and are not interval arithmetic. Boxes are cleared with a margin of 10^{-9} . In the test suite the closed forms agree with 30-digit quadrature at random points with $10^{-4} \leq x \leq 1.5$, $10^{-4} \leq \ell \leq 20$ or $\ell = \infty$ and $10^{-3} \leq \beta\ell \leq 10^3$, which cover the ranges where boxes are evaluated, including the mixture kernels of Table S5, whose largest ratio D_i/T^2 is about 1.25. The largest error found was 1.1×10^{-12} for $x \leq 0.37$ and 2.5×10^{-11} above.

For $0.05 \leq x \leq 0.3$ the bound exceeds a feasible value by at most a relative 3×10^{-4} for $p = 0.9$ and 10^{-3} for $p = 0.9068$. The branch and bound fails only just before the feasibility edges, near $x = 0.370$ for $p = 0.9$, on $0.3585 \leq x \leq 0.3615$ for $p = 0.9036$ and on $0.352 \leq x \leq 0.3545$ for $p = 0.9068$; there Proposition S5 from the last grid point gives $R_{0.9,0.9}(x) \leq 0.132$. Below $x = 0.01$ the lookup also uses Theorem 3 and its $p \geq q$ form, which are within 0.3% of a feasible value there (Fig. S4). At $q = 0.9$ the numerical bounds fall below 1 from $x = 0.013$ on, with the continuity margin $c_{0.9}(5 \times 10^{-4})^{1/2} \approx 4.3 \times 10^{-4}$ keeping them below 1 between grid points, and reach 0.973 at $x = 0.05$.

Exploratory searches

Grid maximization over the extremal family, as in Fig. S1, gives lower bounds of $R_{q,q}$; grid values below 1 do not show that shrinking is safe. This two-dimensional computation over slope and length, and an independent differential-evolution search over the three-parameter family, agree to 1.5×10^{-13} at all 73 values of $x \geq 0.006$

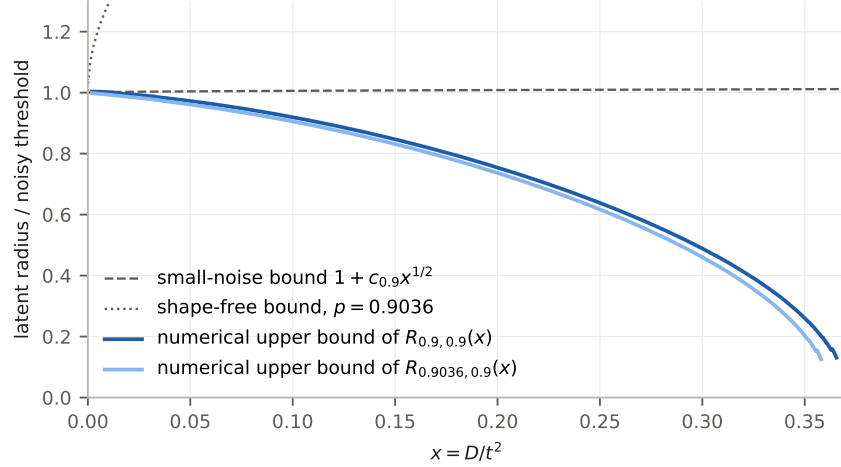


Figure S4: Numerical upper bound of the sharp radius $R_{p,0.9}(x)$ for $p = 0.9$ and for the order-statistic level $p = 0.9036$ ($\delta = 0.04$), with the small-noise bound of Theorem 3 and the shape-free bound $1 + x^{1/2}z_{1-(p-q)/2}$, which follows from $|W| \leq |V| + |e|$.

in `results/r_exact_p0.9_q0.9.csv` (numerical). The search is used only for $x \geq 0.006$: it evaluates the constraint by Gauss–Legendre quadrature, which is inaccurate for steep slopes at small x , and at $x = 0.001$ it returned 1.027, above the bound 1.0006 of Theorem 3.

S6 Heterogeneous and estimated noise variances

Heterogeneous known variances

This section gives the rules of § 4.3, which use the noise variances D_i when they are known, with Gaussian noise. They are an extension: the main results need neither the variances nor the noise law.

The computed radius. Let $T = |V|_{(k)}$ and let

$$\bar{g}(w) = K^{-1} \sum_{i=1}^K \text{pr}(|w + e_i| \leq T)$$

be the average noisy kernel at T . By Proposition S2, with probability at least $1 - \delta$ over calibration samples, $\int \bar{g} dG \geq p_k$, the δ -quantile of the Beta($k, K + 1 - k$) distribution; the proof needs $k \geq K p_k + 1$, which holds for the ranks used here. Let R^{mix} be defined by (2), in units in which the threshold is 1, with the constraint $\int \bar{g}(Tz) dG_T(z) \geq p_k - \omega$ on the law G_T of W/T , where ω bounds the error from compressing the D_i to a few support points. By Proposition S3, which holds for this kernel, the computed radius TR^{mix} covers the latent target with probability at least q , given the data, on an event of probability at least $1 - \delta$, for every log-concave G . This guarantee is for the exact R^{mix} ; the implementation computes a numerical upper bound on it in double precision, not a certificate.

In the heterogeneous rule, ω bounds, for each data set, the sup-norm error of the average kernel after compressing the D_i to 32 support points: the maximum over a grid of spacing $h = 10^{-3}$ plus $Mh^2/8$, with M a bound on the second derivative; it was about 2×10^{-4} . For each data set a numerical upper bound of R^{mix} is computed by the box bounds, which hold term by term for a mixture kernel, and the shape-free bound is used if the branch and bound fails.

Replacing the average kernel by the kernel of a single variance is not safe. With the mean variance plugged in, the radius is 1.4% too short at mean $x = 0.2$ with lognormal spread 1 (numerical). The smallest variance also fails: take $W = b - E_2$ with $b = 1.0435647$ and noise an equal mixture of $N(0, 10^{-8})$ and $N(0, 5 \times 10^{-4})$. The noisy coverage of $[-1, 1]$ exceeds 0.9, while the radius $1 + c_{0.9} 10^{-4}$ given by the smaller variance covers W with probability 0.89977; this was checked in 192-bit ball arithmetic.

A known lower bound on the variances

Corollary S1. *Let $G \in \mathcal{B}$, let the noise be Gaussian, and let $k \geq K p_k + 1$. If $D_i \geq D_{\min}$ for all i and $C_{p',q} < 0$ for some p' with $q \leq p' \leq p_k$, then $s = \max\{0, T + C_{p',q} D_{\min}^{1/2}\}$ satisfies $\Pr_{\mathcal{D}}\{\Pr(|W_{\text{new}}| \leq s \mid \mathcal{D}) \geq q\} \geq 1 - \delta$, whether or not the D_i are otherwise known.*

Proof. On the event of Proposition S2, some i has $\Pr(|W + e_i| \leq T) \geq p_k \geq p'$, and the bound of § 4.3 at $x = D_i/T^2$ gives $Q_q(|W|) \leq T + C_{p',q} D_i^{1/2} \leq T + C_{p',q} D_{\min}^{1/2}$. $\square$

At $q = 0.9$, $K = 110$ and $\delta = 0.05$, where $p_k = 0.906806$, ball arithmetic gives $C_{0.9068,0.9} \leq -0.114$, hence $s = T - 0.114 D_{\min}^{1/2}$, the shrunk threshold of Table S5.

Estimated variances

Suppose the D_i are unknown and estimates $\hat{D}_i$ are available that are independent of the scores, as for the sample means and variances of normal unit data. The order-statistic event of Proposition S2 involves the scores only. Let $\mathcal{A}_\eta$ be an event determined by the $\hat{D}_i$ with $\Pr(\mathcal{A}_\eta) \geq 1 - \eta$. For a vector $D = (D_1, \dots, D_K)$ of variances and a threshold t , let $\bar{g}_{D,t}(w) = K^{-1} \sum_i \Pr(|w + D_i^{1/2} Z| \leq t)$ be the average kernel. For a kernel u in units in which the threshold is 1, let $R_{p,q}[u]$ be the radius (2) with the constraint replaced by $\int u dG \geq p$. A kernel u_t in the original units, at threshold t , becomes $\tilde{u}_t(z) = u_t(tz)$ in these units; in particular $\bar{g}_{D,t}(tz) = \bar{g}_{D/t^2,1}(z)$. Write $R_{p,q}^{\text{mix}}(x_1, \dots, x_K) = R_{p,q}[\bar{g}_{x,1}]$, the radius for the average kernel with variances $x_i = D_i/t^2$ at threshold 1.

Lemma S4. *Let $G \in \mathcal{L}$. Suppose that on $\mathcal{A}_\eta$ a continuous kernel u_t , vanishing at $\pm\infty$, satisfies $u_t \geq \bar{g}_{D,t}$ pointwise for every $t > 0$. Then $s = T R_{p_k,q}[\tilde{u}_T]$, with $\tilde{u}_T(z) = u_T(Tz)$, covers a new latent residual with probability at least q , conditionally on the data, with probability at least $1 - \delta - \eta$. If $\mathcal{D}_\eta$ is a confidence set with $\Pr(D \in \mathcal{D}_\eta) \geq 1 - \eta$, the same holds for $s = T \sup_{d \in \mathcal{D}_\eta} R_{p_k,q}^{\text{mix}}(d/T^2)$. With a compressed kernel, p_k is replaced by $p_k - \omega$ as in § S6.*

Proof. On both events, $\int u_T dG \geq \int \bar{g}_{D,T} dG \geq p_k$. With G_T the law of W/T , which is log-concave, $\int \tilde{u}_T dG_T = \int u_T dG \geq p_k$, so $Q_q(|W|)/T \leq R_{p_k,q}[\tilde{u}_T]$, and Proposition S3 applies to $\tilde{u}_T$. The union bound gives $1 - \delta - \eta$, with no independence between the two events needed. $\square$

Lemma S5. *For $\lambda \geq 1$, $R_{p,q}^{\text{mix}}(\lambda x_1, \dots, \lambda x_K) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x_1, \dots, x_K)$.*

Proof. If W is feasible at λx , then $U = \lambda^{-1/2} W$ satisfies the constraint at threshold $\lambda^{-1/2} \leq 1$, hence at threshold 1, so $Q_q(|W|) = \lambda^{1/2} Q_q(|U|) \leq \lambda^{1/2} R_{p,q}^{\text{mix}}(x)$. $\square$

A confidence box for all K variances needs level $1 - \eta/K$ per area and is useless with one or two degrees of freedom. Survey practice instead models variances through a few parameters, as generalized variance functions do. In the simplest such model $D_i = \sigma^2 a_i$ with known a_i , for instance $a_i = 1/n_i$, and the pooled estimate satisfies $\nu \hat{\sigma}^2 / \sigma^2 \sim \chi_\nu^2$ with $\nu = \sum_i (n_i - 1)$. The confidence set is the segment $\{\sigma^2 a : \ell \leq \sigma^2 \leq h\}$.

Lemma S5 covers the supremum over it by finitely many points without assuming that R is monotone in σ^2 : for $\ell = s_0 < \dots < s_m = h$,

$$\sup_{\ell \leq \sigma^2 \leq h} R_{p_k, q}^{\text{mix}}(\sigma^2 a / T^2) \leq \max_j (s_{j+1} / s_j)^{1/2} U_j, \quad U_j \geq R_{p_k, q}^{\text{mix}}(s_j a / T^2).$$

Plugging in the lower end ℓ alone would not be justified, since R can increase with the noise level (Theorem 3).

In the simulations below most radii are grid values, which are lower bounds of R^{mix} and so carry no guarantee; the rule with numerical upper bounds, as Lemma S4 requires, was run on a subset. Areas have n_i normal units, so $D_i = \sigma_i^2 / n_i$ and the within-area variances are independent of the direct estimates; $K = 110$, mean $D = 0.577$, and 100 replications per setting and law (normal, Laplace, truncated Laplace). The rule uses $\delta = 0.04$ and $\eta = 0.01$, split as 0.009 below and 0.001 above the pooled interval for σ^2 , and needs 18 grid points on average. The radii reported use converged grid values at those points; for the first 30 data sets of every setting and law (270 in all) the rule was rerun with numerical upper bounds U_j at every grid point. The branch and bound succeeded for every data set, and the half-width grew by 0.14% on average and by at most 1.0%, at about 3.6 minutes per data set on one core.

1. With $n_i \in \{2, 3\}$, $\nu \approx 165$, the rule is 6.3–6.5% shorter than noisy conformal calibration and 2.9–3.0% longer than the known-variance rule at the same level, which is 9.0–9.1% shorter than noisy conformal calibration. With $n_i \in \{5, \dots, 10\}$, $\nu \approx 740$, the figures are 7.1–7.7% and 1.6–1.7%. Relative to the known-variance rule at $\delta = 0.05$, which needs no budget for the variances and meets the same final target, 62–63% and 73–74% of its gain survive. Reliability of the rule with grid values was at least 0.98; with numerical upper bounds, on the 30 data sets per setting, it was at least 0.967.
2. The supremum over the interval exceeded the value at its lower end by 0.2%.
3. When the within-area variances differ, with lognormal spread 0.5, the common-variance model is wrong and no guarantee applies; the rule was still reliable, at least 0.99, and 5.9–6.5% shorter than noisy conformal calibration.
4. Without pooling, with each area’s variance estimated from ν own degrees of freedom, a kernel envelope built from area-wise intervals and a binomial bound on the number of misses is 31–40% longer than noisy conformal calibration at $\nu = 2$, within 1.3% of it at $\nu = 50$, and 4.7–5.0% and 6.6–7.2% shorter at $\nu = 200$ and 1000 (30 replications per law; grid values). Whether a better area-wise construction exists is open.

S7 The Markov rule

Proposition S6. *Let $\bar{g}$ be a kernel that is symmetric, continuous, strictly decreasing in $|w|$ and tends to 0 as $|w| \rightarrow \infty$, with $m = \bar{g}(0)$, and suppose that $qm < p < m$. Over all laws G on the real line with $\int \bar{g} dG \geq p$, the supremum of $Q_q(|W|)$ is the radius r that solves $\bar{g}(r) = (p - qm)/(1 - q)$.*

Proof. Let $\tau = (p - qm)/(1 - q) \in (0, m)$. Since $m - \bar{g}(W) \geq 0$, Markov’s inequality gives $\text{pr}\{\bar{g}(W) < \tau\} \leq \{m - p\}/(m - \tau) = 1 - q$, and $\bar{g}(W) < \tau$ exactly when $|W| > r$, so $Q_q(|W|) \leq r$. Conversely, for small $\epsilon > 0$ put mass $q - \epsilon$ at 0 and mass $1 - q + \epsilon$ at the point $w_\epsilon > 0$ with $\bar{g}(w_\epsilon) = \{p - (q - \epsilon)m\}/(1 - q + \epsilon)$. This law meets the constraint with equality, has $Q_q(|W|) = w_\epsilon$, and $w_\epsilon \rightarrow r$ as $\epsilon \rightarrow 0$. $\square$

The average kernel of § S6 satisfies these conditions, so for a fixed rank the Markov rule of Table S5 is the smallest radius that the single order-statistic constraint allows without a shape assumption. With 1 in place of $\bar{g}(0)$ it contains the bound of Guille-Escuret & Ndiaye (2024). It is not a bound on procedures that use several thresholds or tune the rank on independent data, as LatentCP (Zheng et al., 2026) can.

S8 A shifted new residual

Proposition S7. *Let W be log-concave with standard deviation ς , let $\text{pr}(|W| \leq t) \geq q'$ and $|\Delta| \leq \rho\varsigma$. Then $\text{pr}(|W + \Delta| \leq t) \geq q' - \rho$. Conversely, for symmetric unimodal noise, $q_0 > e^{-1}$ and $p = \Psi_{\mathcal{E}}(q_0)$, there are log-concave W_n , with standard deviations ς_n , and noise laws in the class such that, as $n \rightarrow \infty$, the noisy coverage $\text{pr}(|W_n + e_n| \leq t)$ tends to p , the latent coverage $\text{pr}(|W_n| \leq t)$ tends to q_0 , which equals $\lambda_{\mathcal{E}}(p)$, and $\text{pr}(|W_n + \rho\varsigma_n| \leq t) \rightarrow q_0 e^{-\rho}$.*

Proof. A shift by $\Delta > 0$ moves the window $[-t, t]$ to $[-t - \Delta, t - \Delta]$, which loses at most the mass of $(t - \Delta, t]$, at most $\Delta \sup f$. For a log-concave density $\varsigma \sup f \leq 1$, with equality for the exponential law (Bobkov & Chistyakov, 2015), so the loss is at most ρ . For the converse let $b = u^{-1} \log(1/q_0)$ and $W = t + b - E_u$, so that $\varsigma = 1/u$, $\text{pr}(W \leq t) = \text{pr}(E_u \geq b) = q_0$ and $\text{pr}(|W| \leq t) = q_0(1 - e^{-2ut})$, and let the noise be uniform on $[-r/u, r/u]$ with r the maximizer of Theorem 1(i) at q_0 . Take W_n of this form with $u_n t \rightarrow \infty$. Then the left end does not matter in the limit, the latent coverage tends to q_0 and, by scaling, the noisy coverage to $E\{\min(1, q_0 e^{-rU})\} = \Psi_{\mathcal{E}}(q_0) = p$, with U uniform on $[-1, 1]$; a shift by ρ/u gives $\text{pr}(W + \rho/u \leq t) = \text{pr}(E_u \geq b + \rho/u) = q_0 e^{-\rho}$, so the shifted latent coverage tends to $q_0 e^{-\rho}$. The loss $q_0(1 - e^{-\rho}) = q_0 \rho + O(\rho^2)$ shows that the linear order of the bound cannot be improved; it does not attain the coefficient one. $\square$

At $q = 0.9$ the guaranteed latent coverage of noisy coverage 0.9 under symmetric unimodal noise is thus between 0.8881 and 0.8892 for $\rho = 0.01$, and between 0.798 and 0.813 for $\rho = 0.1$. Measured relative to t rather than to ς , a shift has no nontrivial bound: point masses just inside the threshold satisfy every noisy constraint.

S9 Simulations and an application

Table S3 lists the experiments of this section and the claim each supports. The laws of Table S5 and of Table 3 are far from the extremal laws of §§ 2–4, which is why the guarantees are conservative there; the stress test uses an extremal law.

Table S3: Experiments and the claims they support

experiment	claim
rules with the same information	Table 3: width of the ranks of Theorem 2 when the noise is unknown
stress test at the extremal law	Proposition 3: without the raise the latent reliability falls as K grows
rules with known variances	§ S6: the computed radius and the shrunk threshold
analytic rules, 2000 data sets	how conservative the guarantees are for these laws
sensitivity to a shared latent law	the Discussion: the guarantees need one latent law shared by all units
oracle-matched widths	the width needed for a guarantee, separated from the shape of the interval
LatentCP and deconvolution	comparison with the methods of Zheng et al. (2026) and Cohen et al. (2025)
school districts	the rank of Theorem 2, which needs no variances, and rules that plug in estimated variances, which no guarantee covers

Rules with the same information

The design is that of § 5 (experiments/e43_same_information.py), with $q = 0.9$, $\delta = 0.05$, $K = 110$ and 1000, and 2000 data sets in each setting. The latent laws are the normal, Laplace, centred gamma with shape 2 and truncated exponential laws of Table S5, all with mean zero and variance 1. As there, the noise variances are $D_i = 0.577 L_i / E(L_i)$ with L_i lognormal with log-scale 0.7, drawn afresh for each data set, and

no rule is told them. Each noise law is scaled to variance D_i : uniform on $[-(3D_i)^{1/2}, (3D_i)^{1/2}]$, Laplace with scale $(D_i/2)^{1/2}$, t_3 multiplied by $(D_i/3)^{1/2}$, and $D_i^{1/2}(E_1 - 1)$ for the centred exponential. Latent coverage given the data is computed from the latent distribution function. For each setting, the extra width of a rule, in percent, is

$$100\left(\frac{\text{mean width of the rule}}{\text{mean width of the usual rank}} - 1\right).$$

A stress test at the extremal law

`experiments/e38_stress.py` uses the boundary-layer law $W = \theta - x^{1/2}E_u$ with $x = 10^{-4}$. The rate $u = 0.0508$ minimizes the limiting latent coverage $e^{-ub_u(q)}$ at $q = 0.9$; it is not the radius maximizer u^* of § S4. The location θ makes the population noisy coverage $\text{pr}(|W + e| \leq 1)$ equal to 0.9. The noise variances are equal, or lognormal as in Table S5 and scaled to mean x . The population latent coverage is then 0.89918 with equal variances, within 10^{-4} of the certified bound 0.8991 of Corollary 1, so the worst case at this noise level lies in $[0.8991, 0.89918]$, and 0.89959 with unequal ones. The raised rank below is that of Theorem 2 for Gaussian noise. Conditional latent coverage is exact; $\delta = 0.05$; 20000 data sets for $K = 110$ and 1000 and 10000 for $K = 10^4$.

Table S4: Stress test, equal variances: mean conditional latent coverage and reliability (standard errors in parentheses), target 0.9 and 0.95; Table 2 gives the rows for $K = 10^3$ and 10^4 with exact reliabilities

K	rank	k	mean latent coverage	reliability
110	marginal	100	0.9004 (0.0002)	0.531 (0.004)
	usual and raised	105	0.9503 (0.0002)	0.966 (0.001)
1000	marginal	901	0.89931 (0.00007)	0.482 (0.004)
	usual	916	0.91483 (0.00007)	0.943 (0.002)
	raised	917	0.91588 (0.00007)	0.954 (0.001)
10^4	marginal	9001	0.89918 (0.00003)	0.395 (0.005)
	usual	9050	0.90419 (0.00003)	0.922 (0.003)
	raised	9060	0.90522 (0.00003)	0.961 (0.002)

Table S4 and Figure S5 show the conclusion: at this law the usual rank keeps its noisy guarantee but loses the latent one as K grows, and the raised rank of Theorem 2 keeps it. The exact reliability of Proposition 3, $\text{pr}\{\text{Bin}(K, H(r_q)) \leq k - 1\}$ with $H(r_q) = 0.90080$, is 0.943, 0.918, 0.790 and 0.151 for the usual rank at $K = 10^3, 10^4, 10^5$ and 10^6 , against 0.954, 0.958 and, at $K = 10^6$, 0.989 for the raised rank (`experiments/e40_exact_reliability.py`). For every K from 29 to 300 the usual rank's reliability is at least 0.9453, attained at $K = 286$, so its shortfall from 0.95 is below 0.005. The slack is therefore not an artefact of the proofs: without it the latent guarantee fails once the calibration sample is large enough for the order statistic to resolve a coverage difference of 10^{-3} . The effect needs small noise and a latent law with a sharp edge just beyond the threshold, here 2% beyond it.

The simulations agree with the exact values. The usual rank has noisy reliability 0.952 and 0.954 at $K = 1000$ and 10^4 , but misses the latent target by 4 and 10 standard errors. The marginal rank has mean latent coverage below 0.9 once $K \geq 1000$, by 0.0008 at $K = 10^4$, the deficit of Corollary 1, and its exact reliability falls to 0.004 at $K = 10^6$. With unequal variances the reliabilities of the usual and the raised rank are 0.948 and 0.960 at $K = 1000$, and 0.936 and 0.968 at $K = 10^4$. At $K = 110$ the two ranks coincide and the order-statistic level 0.9068 leaves ample room.

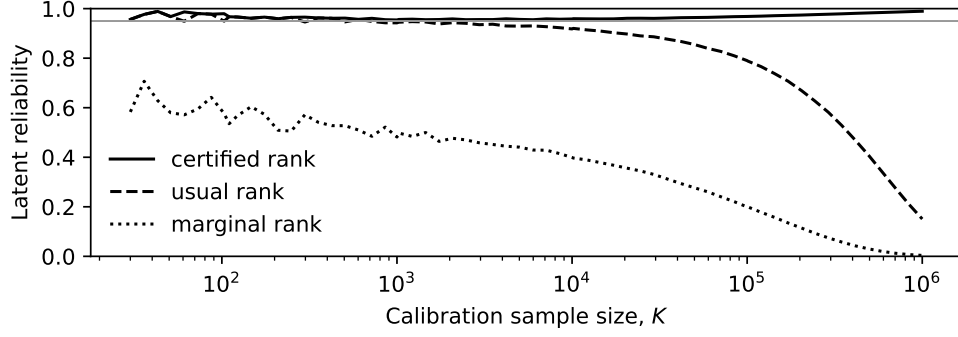


Figure S5: Exact latent reliability at the boundary-layer law of the stress test, equal variances with standard deviation 1% of the threshold, for the raised rank of Theorem 2 (solid, ‘certified rank’), the usual high-probability rank (dashes) and the marginal rank (dots), $\delta = 0.05$; grey, the target 0.95.

Rules with known variances

The rules of § S6, which use the known noise variances, are compared with the noisy threshold, a Fay–Herriot interval and the Markov rule of § S7. The target is $\text{pr}_{\mathcal{D}}\{\text{pr}(W_{\text{new}} \in C \mid \mathcal{D}) \geq 0.9\} \geq 0.95$, with $K = 110$, $\text{var}(W) = 1$ and Gaussian noise with heterogeneous variances $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7. Latent residuals, all with mean zero, are drawn directly; conditional coverage given each data set is exact, and reliability is estimated from 150 data sets per law. Besides four standard log-concave laws we use a truncated exponential law in $\mathcal{F}$, a bimodal law in $\mathcal{B}$ but not in $\mathcal{L}$, and a t_3 law outside $\mathcal{B}$. We compare the noisy threshold at rank 105, the shrunk threshold and the computed radius of § S6, a Fay–Herriot tolerance interval adapted from Chatterjee et al. (2008), and the Markov rule, the radius r with $\bar{g}(r) = \{p_k - q\bar{g}(0)\} / (1 - q)$, the smallest the single constraint allows without a shape assumption (§ S7), at its best rank, 107.

Table S5: Rules with known variances: reliability, times 100, and mean width (in parentheses); target 95

latent law	Markov	Fay–Herriot	noisy threshold	shrunk	computed
normal	100 (5.92)	96 (3.93)	100 (4.96)	100 (4.90)	98 (4.45)
Laplace	100 (6.29)	93 (3.96)	100 (5.15)	100 (5.09)	99 (4.66)
centred gamma	100 (6.16)	100 (3.95)	100 (4.97)	100 (4.90)	100 (4.45)
truncated Laplace	100 (5.89)	85 (3.89)	100 (4.93)	100 (4.87)	97 (4.42)
truncated exponential	100 (6.17)	95 (3.93)	100 (5.06)	100 (4.99)	100 (4.55)
bimodal ($\mathcal{B}$ only)	100 (5.77)	100 (4.01)	100 (4.87)	100 (4.81)	99 (4.35)
t_3 (outside $\mathcal{B}$)	100 (5.70)	97 (3.80)	100 (4.66)	100 (4.60)	98 (4.09)

Noisy threshold, T at rank 105; shrunk, $\max\{0, T - 0.114D_{\min}^{1/2}\}$ with $D_{\min} = \min_i D_i$; computed, TR^{mix} ; centred gamma with shape 2; truncated exponential, density proportional to e^{4w} on $[0, 1]$, standardized; bimodal, the mixture of Lemma S3; widths are full widths. The Markov rule is guaranteed for every law; under the bimodal law the noisy and shrunk thresholds are guaranteed and the computed radius is exploratory; under t_3 the noisy threshold is guaranteed by Anderson’s theorem. Standard errors are at most 2.9 for reliabilities and 0.08 for widths. Reliabilities are rounded: the Fay–Herriot reliability under the truncated exponential law is 0.947, below the target.

The rules with a guarantee reached the target for every law in their class, and the computed radius did so also under the bimodal and t_3 laws, for which it has none. The computed radius was 9.5–12% shorter than the noisy threshold and 24–28% shorter than the Markov rule. The Fay–Herriot interval was 7–15% shorter than the computed radius but fell short of the target under three laws, as detailed below.

Conditional coverage is computed from closed-form latent distribution functions. With 150 data sets the binomial standard error of a reliability is 0.018 at 0.95 and at most 0.029 at the lowest reliability observed, 0.847; for instance $145/150 = 0.967$ has the 95% binomial interval $[0.924, 0.989]$. The computed radius uses the order-statistic level $p_k = 0.9068$ of $\delta = 0.05$. The branch and bound returned a numerical upper bound of R^{mix} for every one of the 1050 data sets, at relative tolerance 10^{-3} in 671, 3×10^{-3} in 363 and 10^{-2} in 16, so the shape-free fallback was never used. With the mean variance in a single Gaussian kernel, which is not a valid plug-in, the numerical radius at level p_k is 2.9–3.2% shorter than at level $p = q$; the order-statistic level thus accounts for about 3 of the 9.5–12 percentage points by which the computed radius is shorter than the noisy threshold.

The Markov rule was run at every fixed rank 106–110; rank 107 is reported because it is best on average over the first four laws, which favours the comparison rule.

The Fay–Herriot interval falls short of the target under the Laplace law by about 1.3 standard errors, under the truncated exponential law, with reliability 0.947, by 0.2, and under the truncated Laplace law, with reliability 0.847, by about 5.8. The Fay–Herriot tolerance interval is our adaptation of the parametric bootstrap of Chatterjee et al. (2008) to an intercept-only model: the bootstrap $(1 - \delta)$ -quantile of the multiplier each refit needs to cover $N(\hat{\mu}, \hat{A})$ with probability 0.9. The latent residuals are drawn directly, with no fitted centre, and the mean noise variance 0.577 matches the ratio of design to model variance in the application below.

Reliability of the analytic rules with 2000 data sets

The noisy threshold, Corollary S1 and the Markov rule need no numerical optimization, so `experiments/e36_analytic_reliability.py` reran them on 2000 fresh data sets for each of the seven laws of Table S5. Every rule reached conditional coverage 0.9 in all 2000 data sets for every law, so each reliability has Clopper–Pearson 95% interval $[0.998, 1]$, including the t_3 law, outside $\mathcal{B}$. Mean conditional coverage was 0.969–0.992 for the noisy threshold, 0.968–0.991 for Corollary S1 and 0.984–0.998 for the Markov rule, and the 5% quantile of conditional coverage was at least 0.947, 0.945 and 0.968.

Sensitivity to a shared latent law

Theorem 2 assumes that the calibration residuals and the new one share a law G , independent of the noise variances. `experiments/e39_sensitivity.py` uses the design of Table S5 with 2000 data sets for the normal, centred gamma and truncated exponential laws and two kinds of departure: residual standard deviation proportional to $D_i^{1/2}$ or $D_i^{-1/2}$, standardized to variance one, with the new area drawing its variance from the same population; and a new area whose residual is shifted by ± 0.2 or ± 0.5 latent standard deviations.

1. Averaged over the new area’s variance, the dispersion departures left the reliability of the noisy threshold and of Corollary S1 at 1, with mean coverage 0.967–0.976, against 0.968–0.984 without departure. Conditionally on the new area’s variance this fails: at the 10% quantile of the variances, when dispersion falls with the variance, and at the 90% quantile, when it rises, mean coverage was 0.918–0.942 and reliability as low as 0.75. Neither case is covered by Theorem 2, which assumes one law independent of the variances: averaged over new areas the reliability was kept in these designs, but not given a new area’s variance.
2. Shifts of ± 0.2 changed mean coverage by at most 0.010, with reliability 1; shifts of ± 0.5 lowered it by up to 0.027, to 0.941 under the truncated exponential law, with reliability at least 0.977 and 5% quantile of conditional coverage 0.908.

3. At the boundary-layer law of the stress test, with $K = 110$ and 20000 data sets, the noisy threshold has reliability 0.969 (0.001) in this run, against 0.966 (0.001) in the independent run of Table S4 and the exact value 0.967; shifting the new residual towards the edge by 0.1 latent standard deviations, 2% of the threshold, lowered mean coverage to 0.860 and reliability to 0.035, and by 0.25 to 0.741 and 0, while shifts away from the edge raised coverage above 0.999.

The robustness in the first two items does not come from the margin between p_k and q ; near an extremal law a systematic difference between sampled and unsampled areas is not tolerated.

Oracle-matched widths

This experiment compares interval shapes at equal reliability. Besides marginal noisy conformal calibration and Fay–Herriot intervals it includes the log-concave rule, the noisy threshold at rank 105 rescaled by the sharp radius $R_{0.9,0.9}$ evaluated at a lower confidence bound for a common noise variance. With seven latent shapes, common and heterogeneous noise and 2000 data sets per setting, each rule’s width was rescaled by a single factor, chosen with knowledge of the latent law, to reach reliability 0.95 exactly. At matched reliability, marginal noisy conformal calibration was shortest in 12 of the 14 settings and the Fay–Herriot tolerance interval in the other 2; relative to noisy conformal calibration the Fay–Herriot tolerance interval was 3.1% shorter to 12.1% longer and the log-concave rule 4.5–11.5% longer (results/conditional_synth_exact_matched.csv). The log-concave rule with its guarantee was a median 15% (range 9–33%) wider than its own oracle-matched width, and the Fay–Herriot tolerance interval a median 6% (range –3 to 16%). So the advantage of the log-concave rule over noisy conformal calibration lies in obtaining a latent guarantee cheaply, not in a better interval shape.

Comparison with LatentCP and deconvolution

experiments/e34_competitors.py runs the following rules on the 600 data sets of the first four laws of Table S5; widths are relative to the computed radius. For the other three laws only the two analytic rules were run.

1. LatentCP (Zheng et al., 2026), with a single level, has a marginal guarantee; in their notation $\alpha = 0.1$ is the latent miscoverage and the noisy conformal threshold is taken at level $1 - \gamma$ with $\gamma = \alpha/2$. Its latent set depends on the new unit’s noise variance, which an unsampled area does not have; we draw it from the distribution that generated the D_i , 400 draws per data set, and average coverage and width over the draws. It was 16–18% longer, with mean coverage 0.976–0.990 and reliability 1.0. The tuned single-level and two-level versions, tuned on half the data, were 15–25% longer. A PAC version with a binomial rank, our adaptation, was 39–58% longer.
2. The deconvolution threshold of Cohen et al. (2025), with the mixture kernel in place of their homoscedastic kernel, collapses at $K = 110$ with their bin width 0.01: most bins are empty and the masked fit gives mean coverage 0.32–0.44. With bin width 0.2 and the empty bins fitted as zeros it had mean coverage 0.88–0.90, reliability 0.45–0.67 and width 22–30% shorter; it has no latent guarantee.
3. The noisy threshold of Theorem 2 was 10.5–11.7% longer and Corollary S1 9.1–10.2% longer, both with reliability 1.0.

The tuning grids of LatentCP and the scan step 0.01 of the deconvolution threshold are our choices; the original papers do not fix them for this setting.

School districts

In the finite population `apipop` (Lumley, 2004), 110 of 325 school districts are used for training and 110 for calibration, with 2 or 3 schools sampled per district. Coverage is the share of the remaining 105 districts that are covered. Design variances are estimated with 1–2 degrees of freedom and plugged in as known by every rule that uses them, a case that no guarantee covers. The noisy threshold at the rank of Theorem 2, $k = 105$ for the three bi-log-concave rows of Table 1, uses none: it had reliability 1.000 in every condition, mean coverage 0.973–0.989, and was 7–12% longer than the heterogeneous rule and 19–26% longer than the Fay–Herriot interval. Its guarantee still needs the noise of the direct estimates to lie in the stated class and the latent residuals to share a bi-log-concave law. Over four conditions of 120 replications, the reliability of the heterogeneous rule, the computed radius with the average kernel, was 0.942–0.992, of the mean-variance rule, which plugs the mean variance into a single Gaussian kernel, 0.983–1.000 and of the Fay–Herriot interval 0.908–0.992, below target in two conditions. The heterogeneous rule at level p_k was 0.8–5.2% shorter than the mean-variance rule calibrated at level $p = q$, mostly because of the level, and 10–16% longer than the Fay–Herriot interval.

S10 Reproducibility

All computations are in the public repository

<https://github.com/kunhailP/latent-coverage-noisy-thresholds>; the submitted version is the tag `biometrika-submission`. The core routines are in `src/uai`: closed forms and the transition coefficient in `extremal.py`, the box bounds in `certify.py`, ball arithmetic in `interval.py`, the noise-class constants and certificates in `noise_classes.py`, and the procedures in `procedures.py` and `estimated.py`. The test suite, run by `make test`, checks the constants, the counterexamples, the centring bounds, the convexity of Ψ for symmetric unimodal noise and the closed forms of Proposition 2, and the `Makefile` lists the command that produces each group of results. Table S6 maps results to scripts in `experiments` and to output files in `results`.

The analytic rules of Theorem 2 and Corollary S1 are `noisy_threshold_halfwidth` and `simple_shrink_halfwidth` in `procedures.py`, and `certified_rank` returns the rank for Gaussian noise; `hetldc_certified` gives the computed radius, while `hetldc_halfwidth` returns a grid value, a lower bound of R^{mix} , and is not a guaranteed radius.

S11 An open problem: the extremal log-concave noise law

This section is not used by any result. It records a partial argument towards the conjecture that the centred exponential law attains $\Psi_{\mathcal{E}}$ for mean-zero log-concave noise, which Theorem 1(ii) leaves open; the certified bounds there do not depend on it.

Since $h(y) = qe^{-\max(y, \ell)}$, with $X = -\epsilon$ and $c = -\ell$,

$$\Psi_{\mathcal{E}}(q) = q \sup E\{e^{\min(X, c)}\},$$

the supremum over mean-zero log-concave X , and the conjectured extremal law is $X = \sigma(1 - E_1)$, with E_1 standard exponential. Laws bounded above by c are covered analytically. If $X \leq s$ almost surely with $0 < s \leq c$, the log-density of X minus the affine log-density of $s(1 - E_1)$ is concave on $(-\infty, s]$, so the difference of densities changes sign at most twice; with equal masses and means, the difference of distribution functions changes sign once, from negative to positive, so X is smaller than $s(1 - E_1)$ in the convex order. As $x \mapsto e^{\min(x, c)}$ is convex on $(-\infty, s]$, $E\{e^{\min(X, c)}\} \leq E\{e^{\min(s(1 - E_1), c)}\}$. The open case is a law with mass above c , where the objective is not convex; the extremal law lies there, with $\sigma = 0.11777 > c = 0.10536$

Table S6: Results, scripts and output files

result	script	output
Theorem 1, noise classes	e41.noise.classes.py	noise.classes.json
Proposition 2, unimodal latent laws	e42.unimodal.boundary.py	printed
Table 3, rules with the same information	e43.same.information.py	same.information.csv
Lemma S1, Corollary 1: $c_{p,q}$, $C_{p,q}$, slack	e27.interval.constants.py	interval.constants.json
Coverage form of Corollary 1	e37.coverage.transfer.py	coverage.transfer.json
Numerical upper bounds of the radius	e24.certified.table.py, e25.certified ldc.py	certified.R.csv, certified_ ldc.summary.csv
Centring and transition values, Fig. S2	e32.centering.py, fig_ transition.py	centering.json
Location of the mean, Fig. S3	e35.mean.location.py, fig_ mean.location.py	mean.location.csv
Table S5, rules with known variances	e20.hetldc.synth.py, e31_ shape.free.py	hetldc.synth.summary.csv, shape.free.summary.csv
Analytic rules, 2000 data sets	e36.analytic_ reliability.py	analytic.reliability_ summary.csv
Stress test, Table 2 and Table S4	e38.stress.py	stress.summary.csv
Proposition 3, Fig. S5	e40.exact.reliability.py	exact.reliability.csv
Sensitivity to a shared latent law	e39.sensitivity.py	sensitivity.summary.csv, sensitivity.stress.csv
Oracle-matched widths	e15.e12.exact.py	conditional.synth.exact_ matched.csv
LatentCP and deconvolution	e34.competitors.py	competitors.summary.csv
Estimated variances	e29.estimated.scale.py, e30.areawise.variance.py	estimated.scale.summary.csv, areawise.variance.summary.csv
School districts	e21.hetldc.apipop.py	hetldc.apipop.summary.csv
Shape-free rule	e31.shape.free.py	shape.free.summary.csv
Lemma S3, bimodal law	uai.interval.bimodal.blc_ certificate	test suite

at $q = 0.9$, and choosing the scale of the comparison law freely gives crossing patterns from which no sign follows.

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